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Data Structures

Dsu

```

/* DSU (Disjoint Set Union / Union-Find)
 * Mantem conjuntos disjuntos com uniao e busca eficiente.
 * Complexidade: O(a(N)); a e a inversa de Ackermann.
 * Memoria: O(N)
 * Metodos:
 * - find(i): Retorna o representante do conjunto de i.
 * - join(i, j): Une os conjuntos de i e j, retorna true se uniu.
 * - same(i, j): Verifica se i e j estao no mesmo conjunto.
 * - size(i): Retorna o tamanho do conjunto de i.
 */
056 struct DSU {
1A8     int n;
923     vector<int> parent;
10F     vector<int> sz;
75E     int num_sets;

87D     DSU(int n) : n(n), parent(n), sz(n, 1), num_sets(n) {
4D6         iota(parent.begin(), parent.end(), 0);
3FC     }

C42     int find(int i) {
331         if (parent[i] == i) return i;
         return parent[i] = find(parent[i]);
565     }
72C }

058     bool join(int i, int j) {
477         i = find(i), j = find(j);
41E         if (i != j) {
4C2             if (sz[i] < sz[j]) swap(i, j);
1F0             parent[j] = i;
529             sz[i] += sz[j];
CA8             num_sets--;
8A6             return true;
20F         }
D1F         return false;

```

```

30C     }
00C     int size(int i) { return sz[find(i)]; }
DA3     bool same(int i, int j) { return find(i) == find(j); }
EF5 };

```

Dsu Rollback

```

/* DSU com Rollback (Union by Size, sem path compression)
 * Complexidade: O(log N) para find, join e rollback.
 * Memoria: O(N + calls a join)
 * Metodos:
 * - find(i): raiz do conjunto de i.
 * - join(i, j): une os conjuntos; sempre empilha no historico
   (mesmo se ja estavam unidos). Retorna true se uniu.
 * - rollback(): desfaz o ultimo join (empilhado ou fantasma).
 * - same(i, j): verifica se i e j estao no mesmo conjunto.
 * - size(i): tamanho do conjunto de i.
 *
 * IMPORTANTE: rollback() deve ser chamado uma vez por join(),
 * independente do retorno - join() sempre empilha no historico.
 */
DE6 struct DSURollback {
ADC     int n, num_sets;
89E     vector<int> parent, sz;
AD4     vector<pair<int,int>> history;

A31     DSURollback(int n) : n(n), num_sets(n), parent(n), sz(n, 1) {
4D6         iota(parent.begin(), parent.end(), 0);
F06     }

C42     int find(int i) {
7B9         while (parent[i] != i) i = parent[i];
D9A         return i;
4BA     }

D58     bool join(int i, int j) {
650         i = find(i); j = find(j);
426         if (i == j) { history.push_back({-1, -1}); return false; }
4C2         if (sz[i] < sz[j]) swap(i, j);
1F0         parent[j] = i;
529         sz[i] += sz[j];
CA8         num_sets--;
0E6         history.push_back({j, i});
8A6         return true;
480     }

5CF     void rollback() {
F07         auto [child, p] = history.back();
7D1         history.pop_back();
D1B         if (child != -1) {
B95             sz[p] -= sz[child];
B1B             parent[child] = child;
F4E             num_sets++;
708         }
0D9     }

00C     int size(int i) { return sz[find(i)]; }
DA3     bool same(int i, int j) { return find(i) == find(j); }
FA9 };

```

Mo

```

/* Mo's Algorithm - O((N + Q) * sqrt(N))
 * queries: vetor de {l, r} 0-indexado, fechado.
 * add(i), rem(i): adiciona/removem o elemento i da janela.
 * ans(qi): registra resposta para a query de indice qi.
 *
 * Exemplo:
 * vector<int> ans(q);
 * Mo mo(n, queries);
 * mo.run(
 *     [&](int i) { ... },
 *     [&](int i) { ... },
 *     [&](int qi) { ans[qi] = ...; }
 * );

```

```

*/
0DF struct Mo {
7F4   int n, block;
670   struct Query {
738     int l, r, idx;
304   };
240   vector<Query> queries;

8CA   Mo(int n, vector<pair<int, int>> &q)
10A     : n(n), block(max(1, (int)sqrt(n))) {
D26     for (int i = 0; i < (int)qs.size(); i++)
9AA       queries.push_back({qs[i].first, qs[i].second, i});
F94     sort(queries.begin(), queries.end(),
724       [&](const Query &a, const Query &b) {
ED6         int ba = a.l / block, bb = b.l / block;
A2D         if (ba != bb) return ba < bb;
FDF         return (ba & 1) ? a.r > b.r : a.r < b.r;
FC3       });
8F7   }

D5A   template <typename Add, typename Rem, typename Ans>
743   void run(Add add, Rem rem, Ans ans) {
73A     int cur_l = 0, cur_r = -1;
FFE     for (auto &[l, r, idx] : queries) {
90C       while (cur_r < r) add(++cur_r);
6A3       while (cur_l > l) add(--cur_l);
1B1       while (cur_r > r) rem(cur_r--);
E02       while (cur_l < l) rem(cur_l++);
414       ans[idx];
5B1     }
8B8   }
1FF };

/* HilbertMo – Mo com ordenacao pela curva de Hilbert
* Mesma interface que Mo, mas com constante menor na pratica.
*/
CD0 struct HilbertMo {
670   struct Query {
738     int l, r, idx;
BFE     ll ord;
C8D   };
240   vector<Query> queries;

D16   static ll hilbert(int x, int y, int n) {
B72     ll d = 0;
F8A     for (int s = n >> 1; s; s >= 1) {
7B9       int rx = (x & s) > 0, ry = (y & s) > 0;
71B       d += (ll)s * s * ((3 * rx) ^ ry);
069       if (!ry) {
1C6         if (rx) {
731           x = s - 1 - x;
D41           y = s - 1 - y;
1FD         }
9DD         swap(x, y);
FFF       }
0D1     }
BE2     return d;
416   }

E76   HilbertMo(int n, vector<pair<int, int>> &q) {
ABE     int hn = 1;
FCC     while (hn < n) hn <= 1;
D26     for (int i = 0; i < (int)qs.size(); i++)
082       queries.push_back(
A55         {qs[i].first, qs[i].second, i,
979           hilbert(qs[i].first, qs[i].second, hn)});
F94     sort(queries.begin(), queries.end(),
B18       [&](const Query &a, const Query &b) {
73A         return a.ord < b.ord;
A3A       });
0D6   }

D5A   template <typename Add, typename Rem, typename Ans>
743   void run(Add add, Rem rem, Ans ans) {
73A     int cur_l = 0, cur_r = -1;

```

```

277   for (auto &[l, r, idx, ord] : queries) {
90C     while (cur_r < r) add(++cur_r);
6A3     while (cur_l > l) add(--cur_l);
1B1     while (cur_r > r) rem(cur_r--);
E02     while (cur_l < l) rem(cur_l++);
414     ans[idx];
073   }
DC2   }
1AD };

Parallel Binary Search

/* Parallel Binary Search – O((N + Q) log Q * op)
* Para cada entidade em [0, N), encontra o primeiro evento
* em [0, Q) apos o qual uma condicao monotona e satisfeita.
* Requer que a condicao seja monotona: uma vez satisfeita,
* permanece.
*
* apl(i): aplica o evento i na estrutura de dados.
* rlb(i): desfaz o evento i da estrutura de dados.
* chk(i): retorna true se a entidade i satisfaz a
*         condicao dado o estado atual da estrutura.
*
* ans[i] = primeiro evento que satisfaz, ou -1 se nunca.
*
* Exemplo (soma dos setores de i >= limite[i]):
*   ParallelBS pbs(n, q);
*   pbs.run(
*     [&](int i) { seg.update(l[i], r[i], {x[i]}); },
*     [&](int i) { seg.update(l[i], r[i], {-x[i]}); },
*     [&](int idx) {
*       ll sum = 0;
*       for (int s : sectors[idx]) {
*         sum += seg.query(s, s.val);
*         if (sum >= lim[idx]) return true;
*       }
*       return false;
*     }
*   );
*   for (int i = 0; i < n; i++)
*     cout << (pbs.ans[i] != -1 ? pbs.ans[i]+1 : -1);
*/

540 struct ParallelBS {
3B9   int n, q;
DAB   vector<int> ans;

DC7   ParallelBS(int n, int q) : n(n), q(q), ans(n, -1) {}

8BC   template <typename Apl, typename Rlb, typename Chk>
708   void run(Apl apl, Rlb rlb, Chk chk) {
56A     vector<int> actv(n);
A0A     iota(actv.begin(), actv.end(), 0);
483     solve(0, q - 1, actv, apl, rlb, chk);
41B   }

BF2 private:
8BC   template <typename Apl, typename Rlb, typename Chk>
EF4   void solve(int l, int r, vector<int>& actv,
9A3     Apl apl, Rlb rlb, Chk chk) {
658     if (actv.empty()) return;

893     if (l == r) {
A2E       apl(l);
E01       for (auto x : actv) if (chk(x)) ans[x] = l;
623       rlb(l);
505       return;
78E     }

318     int mid = l + (r - l) / 2;
9EA     for (int i = l; i <= mid; i++) apl(i);

37F     vector<int> pass, not_pass;
3FE     for (auto x : actv)
C9B       (chk(x) ? pass : not_pass).push_back(x);

9F1     solve(mid + 1, r, not_pass, apl, rlb, chk);

```

```

A2B     for (int i = l; i <= mid; i++) rlb(i);
50E     solve(l, mid, pass, apl, rlb, chk);
3FC   }
616 };

```

Fenwick Tree

```

/* Fenwick Tree (Point Update, Prefix Query)
* Realiza atualizacoes pontuais e consultas de prefixo.
* Complexidade: O(log N) para update e query.
* Memoria: O(N)
* Requisitos:
* - NODE: operator+=, operator- e construtor default.
*/

/* --- Exemplo de NODE (Soma) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    void operator+=(const Node& other) { val += other.val; }
    Node operator-(const Node& other) const {
        return Node(val - other.val);
    }
};
*/

8A0 template <typename NODE>
0F1 struct FenwickTree {
060   int N;
9B8   vector<NODE> bit;

5D2   FenwickTree(int n) : N(n), bit(n + 1) {}

D6B   void update(int idx, NODE v) {
B2F     for (idx++; idx <= N; idx += idx & -idx) {
046       bit[idx] += v;
23E     }
9DA   }

FCA   NODE query(int idx) { // [0, idx]
ADA     NODE res;
159     for (idx++; idx > 0; idx -= idx & -idx) {
8E6       res += bit[idx];
A38     }
B50     return res;
C00   }

762   NODE query(int l, int r) { // [l, r]
7C3     if (l > r) return NODE();
A4E     return query(r) - query(l - 1);
821   }
134 };

Fenwick Tree 2D

/* Fenwick Tree 2D (Point Update, Rectangle Query)
* Atualizacoes pontuais e consultas de prefixo em matrizes.
* Complexidade: O(log N * log M) para update e query.
* Memoria: O(N * M)
* Requisitos:
* - NODE: operator+=, operator- e construtor default.
*/

/* --- Exemplo de NODE (Soma) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    void operator+=(const Node& other) { val += other.val; }
    Node operator-(const Node& other) const {
        return Node(val - other.val);
    }
};
*/

8A0 template <typename NODE>
2E6 struct FenwickTree2D {
610   int N, M;
803   vector<vector<NODE>> bit;

```

```

265 FenwickTree2D(int n, int m)
B1E : N(n), M(m), bit(n + 1, vector<NODE>(m + 1)) {}

274 void update(int r, int c, NODE v) { // point update
3BA for (int i = r + 1; i <= N; i += i & -i) {
786 for (int j = c + 1; j <= M; j += j & -j) {
9ED bit[i][j] += v;
95F }
A1F }
BF9 }

82E NODE query(int r, int c) { // [(0,0), (r,c)]
ADA NODE res;
8D9 for (int i = r + 1; i > 0; i -= i & -i) {
83F for (int j = c + 1; j > 0; j -= j & -j) {
01D res += bit[i][j];
CB5 }
E20 }
B50 return res;
943 }

// Retangulo fechado [(r1,c1), (r2,c2)].
05F NODE query(int r1, int c1, int r2, int c2) {
5C4 if (r1 > r2 || c1 > c2) return NODE();
10A return query(r2, c2) - query(r1 - 1, c2) -
7F5 query(r2, c1 - 1) + query(r1 - 1, c1 - 1);
0E7 }
FF7 };

```

Dynamic Fenwick Tree 2D

```

/* Fenwick Tree 2D Sparse (Point Update, Rectangle Query)
* Suporta coordenadas ate 10^9 usando compressao interna.
* Complexidade: Build O(P log P), Update/Query O(log^2 P).
* Memoria: O(P log P).
* Requisitos:
* - NODE: operator+=, operator-, operator+ e construtor default.
* - Offline: passe todos os pontos de interesse no build.
*
* Creditos: Adaptado de TFG (Thiago Ferreira Goncalves)
* Fonte: github.com/tfg50/Competitive-Programming
*/

/* --- Exemplo de NODE (Soma com Inclusao-Exclusao) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}

    void operator+=(const Node& other) {
        val += other.val;
    }

    Node operator-(const Node& other) const {
        return Node(val - other.val);
    }

    Node operator+(const Node& other) const {
        return Node(val + other.val);
    }
};
*/

8A0 template <typename NODE>
222 struct FenwickTree2DSparse {
6C1 vector<int> ord;
803 vector<vector<NODE>> bit;
E76 vector<vector<int>> coord;

BEF FenwickTree2DSparse(vector<pii> pts) {
CD7 sort(pts.begin(), pts.end());
5A8 for (auto [x, y] : pts) {
DC4 if (ord.empty() || x != ord.back()) ord.push_back(x);
ABA }

261 bit.resize(ord.size() + 1);
3EB coord.resize(ord.size() + 1);

A0A sort(pts.begin(), pts.end(), [](const pii &a, const pii &b) {
463 return a.second < b.second;

```

```

19C });
5A8 for (auto [x, y] : pts) {
9DE for (int i = get_x(x); i < bit.size(); i += i & -i) {
3E1 if (coord[i].empty() || coord[i].back() != y)
739 coord[i].push_back(y);
4F2 }
B39 }

8C2 for (int i = 0; i < bit.size(); i++)
F3E bit[i].assign(coord[i].size() + 1, NODE());
D9D }

7E8 int get_x(int x) {
2C8 return upper_bound(ord.begin(), ord.end(), x) - ord.begin();
95C }

D5E int get_y(int i, int y) {
E18 auto &c = coord[i];
BBD return upper_bound(c.begin(), c.end(), y) - c.begin();
418 }

5CF void update(int x, int y, NODE v) {
9DE for (int i = get_x(x); i < bit.size(); i += i & -i) {
813 for (int j = get_y(i, y); j < bit[i].size(); j += j & -j) {
9ED bit[i][j] += v;
34E }
F86 }
390 }

84C NODE query(int x, int y) {
ADA NODE res;
FAD for (int i = get_x(x); i > 0; i -= i & -i) {
263 for (int j = get_y(i, y); j > 0; j -= j & -j) {
01D res += bit[i][j];
359 }
92F }
B50 return res;
7C4 }

FCA NODE query(int x1, int y1, int x2, int y2) {
276 return query(x2, y2) - query(x1 - 1, y2) -
12D query(x2, y1 - 1) + query(x1 - 1, y1 - 1);
944 }
B5F };

```

Sparse Table

```

/* Sparse Table (Static RMQ)
* Realiza consultas em range em tempo constante.
* Complexidade: O(N log N) no build, O(1) na query.
* Memoria: O(N log N)
* Requisitos:
* - NODE deve ter: static merge(const NODE&, const NODE&) e
* construtor.
* - A operacao DEVE ser idempotente: f(a, a) = a.
*/

/* --- Exemplo de NODE (Indice do Minimo) ---
struct Node {
    int val, pos;
    Node(int v = 2e9, int p = -1) : val(v), pos(p) {}

    static Node merge(const Node& l, const Node& r) {
        return l.val <= r.val ? l : r;
    }
};

// Uso: SparseTable<Node> st(v) onde v e vector<Node>
// for (int i = 0; i < n; i++) {
//     cin >> x; v[i] = Node(x, i);
// }
*/

8A0 template <typename NODE>
7E9 struct SparseTable {
5F0 int N, K;
6F5 vector<vector<NODE>> st;
4A7 vector<int> lg;

```

```

67A template <typename T>
E4A SparseTable(const vector<T> &v) : N(v.size()) {
A77 K = (N > 0) ? __lg(N) : 0;
3A9 st.assign(K + 1, vector<NODE>(N));
641 lg.assign(N + 1, 0);
BB5 for (int i = 2; i <= N; i++) lg[i] = lg[i / 2] + 1;
DDD for (int i = 0; i < N; i++) st[0][i] = NODE(v[i]);
2CE for (int j = 1; j <= K; j++)
285 for (int i = 0; i + (1 << j) <= N; i++)
4E1 st[j][i] = NODE::merge(st[j - 1][i],
A64 st[j - 1][i + (1 << (j - 1))]);
CE2 }

762 NODE query(int l, int r) {
2C3 int j = lg[r - l + 1];
8F2 return NODE::merge(st[j][l], st[j][r - (1 << j) + 1]);
843 }
231 };

```

Disjoint Sparse Table

```

/* Disjoint Sparse Table (Static Range Query)
* Query O(1) para qualquer operacao associativa.
* Complexidade: O(N log N) no build, O(1) na query.
* Memoria: O(N log N)
* Requisitos:
* - NODE deve ter: static merge(const NODE&, const NODE&) e
* construtor.
* - Operacao deve ser ASSOCIATIVA: f(f(a,b),c) =
* f(a,f(b,c)).
*/

/* --- Exemplo de NODE (Soma em Range) ---
struct Node {
    ll val;
    Node(ll v = 0) : val(v) {}
    static Node merge(const Node& l, const Node& r) {
        return Node(l.val + r.val);
    }
};
*/

8A0 template <typename NODE>
906 struct DisjointSparseTable {
5F0 int N, K;
6F5 vector<vector<NODE>> st;

67A template <typename T>
68E DisjointSparseTable(const vector<T> &v) : N(v.size()) {
3AA if (N == 0) return;
475 K = __lg(2 * N - 1);
C78 st.assign(K, vector<NODE>(1 << K));

DDD for (int i = 0; i < N; i++) st[0][i] = NODE(v[i]);
C54 for (int j = 1; j < K; j++) {
1C0 int len = 1 << j;
C2C for (int m = len; m <= (1 << K); m += 2 * len) {
A34 if (m < N) st[j][m] = st[0][m];
6D6 for (int i = m + 1; i < min(N, m + len); i++)
0EA st[j][i] = NODE::merge(st[j][i - 1], st[0][i]);
C2E if (m - 1 < N) st[j][m - 1] = st[0][m - 1];
226 for (int i = m - 2; i >= m - len; i--)
604 st[j][i] = NODE::merge(st[0][i], st[j][i + 1]);
643 }
3F0 }
4D7 }

762 NODE query(int l, int r) {
434 if (l == r) return st[0][l];
B89 int j = __lg(l ^ r);
894 return NODE::merge(st[j][l], st[j][r]);
33A }
04C };

```

Segment Tree

```

/* Segment Tree (Point Update, Range Query)
 * Complexidade: O(log N) para update e query.
 * Memoria: O(4*N)
 * Requisitos:
 * - NODE deve ter: static merge(L, R), apply(V), e
 * construtor identidade.
 */

/* --- Exemplo de NODE (Soma com Update de Substituicao) ---
struct Node {
    ll val = 0;
    Node(ll v = 0) : val(v) {}

    static Node merge(const Node& l, const Node& r) {
        return Node(l.val + r.val);
    }

    // Para Point Set: val = v;
    // Para Point Add: val += v;
    void apply(ll v) {
        val = v;
    }
};
*/

8A0 template <typename NODE>
383 struct SegTree {
060     int N;
B1C     vector<NODE> seg;

FDE     explicit SegTree(int n) : N(n), seg(4 * n) {}

67A     template <typename T>
5DE     SegTree(const vector<T> &v) : SegTree((int)v.size()) {
231         build(1, 0, N - 1, v);
73D     }

67A     template <typename T>
5B9     void build(int no, int l, int r, const vector<T> &v) {
893         if (l == r) {
C3E             seg[no] = NODE(v[l]);
505             return;
B2E         }
27E         int m = (l + r) >> 1;
35E         build(no << 1, l, m, v);
C55         build((no << 1) | 1, m + 1, r, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349     }

637     template <typename V>
AF7     void update(int no, int l, int r, int idx, V val) {
893         if (l == r) {
D88             seg[no].apply(val);
505             return;
EC9         }
27E         int m = (l + r) >> 1;
B73         if (idx <= m) update(no << 1, l, m, idx, val);
99C         else update((no << 1) | 1, m + 1, r, idx, val);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
78C     }

38D     NODE query(int no, int l, int r, int a, int b) {
2E6         if (b < l || r < a) return NODE();
83F         if (a <= l && r <= b) return seg[no];
27E         int m = (l + r) >> 1;
AFE         return NODE::merge(query(no << 1, l, m, a, b),
074             query((no << 1) | 1, m + 1, r, a, b));
365     }

637     template <typename V>
864     void update(int idx, V val) {
E1B         update(1, 0, N - 1, idx, val);
22F     }

36F     NODE query(int l, int r) { return query(1, 0, N - 1, l, r); }
0C7 };

```

Lazy Segment Tree

```

/* Lazy Segment Tree (Range Update, Range Query)
 * Realiza atualizacoes e consultas em range.
 * Complexidade: O(log N) para update e query.
 * Memoria: O(4*N)
 * Requisitos:
 * - NODE deve ter: static merge(L, R), apply(TAG, L, R) e
 * construtor identidade.
 * - TAG deve ter: compose(TAG) e construtor identidade.
 */

/* --- Exemplo de NODE e TAG (Soma e Multiplicacao com
 * Modulo) ---
struct Tag {
    ll mul = 1, add = 0;
    void compose(const Tag& t) {
        add = (add * t.mul + t.add) % MOD;
        mul = (mul * t.mul) % MOD;
    }
};

struct Node {
    ll val = 0;
    Node(ll v = 0) : val(v) {}
    static Node merge(const Node& l, const Node& r) {
        return Node((l.val + r.val) % MOD);
    }
    void apply(const Tag& t, int l, int r) {
        val = (val * t.mul + t.add * (r - l + 1)) % MOD;
    }
};
*/

708 template <typename NODE, typename TAG>
2BA struct LazySegmentTree {
060     int N;
B1C     vector<NODE> seg;
71A     vector<TAG> lazy;

BB8     explicit LazySegmentTree(int n)
63C         : N(n), seg(4 * n), lazy(4 * n) {}

67A     template <typename T>
1BA     LazySegmentTree(const vector<T> &v)
002         : LazySegmentTree((int)v.size()) {
231         build(1, 0, N - 1, v);
9BB     }

67A     template <typename T>
5B9     void build(int no, int l, int r, const vector<T> &v) {
893         if (l == r) {
C3E             seg[no] = NODE(v[l]);
505             return;
B2E         }
27E         int m = (l + r) >> 1;
35E         build(no << 1, l, m, v);
C55         build((no << 1) | 1, m + 1, r, v);
DEB         seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349     }

F1A     void push(int no, int lo, int hi) {
946         int m = (lo + hi) >> 1;
8F3         int l = no << 1, r = l | 1;

D09         seg[l].apply(lazy[no], lo, m);
9FE         lazy[l].compose(lazy[no]);

204         seg[r].apply(lazy[no], m + 1, hi);
3FC         lazy[r].compose(lazy[no]);

47F         lazy[no] = TAG();
DD1     }

130     void update(int no, int l, int r, int a, int b, const TAG &v) {
2E6         if (b < l || r < a) return;
02B         if (a <= l && r <= b) {
8C7             seg[no].apply(v, l, r);
92C             lazy[no].compose(v);

```

```

505         return;
81F     }
FF9     push(no, l, r);
27E     int m = (l + r) >> 1;
9AB     update(no << 1, l, m, a, b, v);
087     update((no << 1) | 1, m + 1, r, a, b, v);
DEB     seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
A1A }

38D     NODE query(int no, int l, int r, int a, int b) {
2E6         if (b < l || r < a) return NODE();
83F         if (a <= l && r <= b) return seg[no];
FF9         push(no, l, r);
27E         int m = (l + r) >> 1;
AFE         return NODE::merge(query(no << 1, l, m, a, b),
074             query((no << 1) | 1, m + 1, r, a, b));
80A     }

4E7     void update(int l, int r, const TAG &v) {
E27         update(1, 0, N - 1, l, r, v);
0D7     }
36F     NODE query(int l, int r) { return query(1, 0, N - 1, l, r); }
3CE };

```

Dual Segment Tree

```

/* Dual Segment Tree (Range Update, Point Query)
 * Realiza atualizacoes em range e consultas pontuais.
 * Complexidade: O(log N) para update e get.
 * Memoria: O(4*N)
 * Requisitos:
 * - TAG deve ter: compose(TAG), apply(T&) e construtor
 * identidade.
 * - T e o tipo do dado nas folhas.
 * Exemplo de uso: DualSegTree<ll, MyTag> st(v);
 */

/* --- Exemplo de TAG (Soma e Multiplicacao com Modulo) ---
struct Tag {
    ll mul = 1, add = 0;
    void compose(const Tag& t) {
        mul = (mul * t.mul) % MOD;
        add = (add * t.mul + t.add) % MOD;
    }
    void apply(ll& val) { val = (val * mul + add) % MOD; }
};
*/

2E5 template <typename T, typename TAG>
9C1 struct DualSegTree {
060     int N;
71A     vector<TAG> lazy;
E4B     vector<T> leaves;

F06     explicit DualSegTree(int n) : N(n), lazy(4 * n), leaves(n) {}
5F6     DualSegTree(const vector<T> &v)
CE6         : N((int)v.size()), lazy(4 * v.size()), leaves(v) {}

78E     void push(int no) {
8A1         int l = no << 1;
C9C         int r = l | 1;
9FE         lazy[l].compose(lazy[no]);
3FC         lazy[r].compose(lazy[no]);
47F         lazy[no] = TAG();
283     }

130     void update(int no, int l, int r, int a, int b, const TAG &v) {
2E6         if (b < l || r < a) return;
02B         if (a <= l && r <= b) {
92C             lazy[no].compose(v);
505             return;
CA0         }
523         push(no);
27E         int m = (l + r) >> 1;
9AB         update(no << 1, l, m, a, b, v);
087         update((no << 1) | 1, m + 1, r, a, b, v);
BEB     }

```

```

629 T get(int no, int l, int r, int idx) {
893   if (l == r) {
405     T res = leaves[idx];
2E3     lazy[no].apply(res);
850     return res;
BE7   }
523   push(no);
27E   int m = (l + r) >> 1;
52E   if (idx <= m) return get(no << 1, l, m, idx);
483   else return get((no << 1) | 1, m + 1, r, idx);
C4A }

C8C void update(int l, int r, const TAG &t) {
56D   update(1, 0, N - 1, l, r, t);
A21 }
AE4 T get(int idx) { return get(1, 0, N - 1, idx); }
25C };

```

Segment Tree Beats

```

/* Segment Tree Beats (Range Update, Range Query)
 * Lazy Seg Tree com condicoes de parada e aplicacao de tag.
 * Complexidade amortizada:  $O(N \log^2 N)$  por sequencia de updates.
 * Memoria:  $O(4*N)$ 
 * Requisitos:
 * - NODE deve ter: static merge(L, R), apply(TAG, L, R),
 *   break_condition(TAG) e tag_condition(TAG),
 *   e construtor identidade.
 * - TAG deve ter: compose(TAG) e construtor identidade.
 *
 * break_condition(tag): ignora update ja satisfeito (mx <= v).
 * tag_condition(tag): aplica sem descer (mx2 < v <= mx).
 */

/* --- Exemplo de NODE e TAG (chmin em range + query de
 * soma/max) ---
struct Tag {
    ll val = LLONG_MAX; // identidade: nao faz nada
    void compose(const Tag& t) {
        val = min(val, t.val);
    }
};

struct Node {
    ll sum, mx, mx2;
    int cnt_mx, sz;
    Node()
        : sum(0), mx(LLONG_MIN), mx2(LLONG_MIN),
          cnt_mx(0), sz(0) {}
    Node(ll v)
        : sum(v), mx(v), mx2(LLONG_MIN), cnt_mx(1),
          sz(1) {}
    static Node merge(const Node& l, const Node& r) {
        Node res;
        res.sz = l.sz + r.sz;
        res.sum = l.sum + r.sum;
        if (l.mx == r.mx) {
            res.mx = l.mx; res.cnt_mx = l.cnt_mx + r.cnt_mx;
            res.mx2 = max(l.mx2, r.mx2);
        } else if (l.mx > r.mx) {
            res.mx = l.mx; res.cnt_mx = l.cnt_mx;
            res.mx2 = max(l.mx2, r.mx);
        } else {
            res.mx = r.mx; res.cnt_mx = r.cnt_mx;
            res.mx2 = max(l.mx, r.mx2);
        }
        return res;
    }
};

// aplica a tag quando tag_condition e verdadeiro
void apply(const Tag& t, int l, int r) {
    sum -= (ll)(mx - t.val) * cnt_mx;
    mx = t.val;
}

```

```

// ignora o update completamente (no ja satisfaz)
bool break_condition(const Tag& t) const {
    return mx <= t.val;
}

// pode aplicar a tag diretamente sem descer
bool tag_condition(const Tag& t) const {
    return mx2 < t.val;
}

};
*/

708 template <typename NODE, typename TAG>
BC8 struct SegTreeBeats {
060   int N;
B1C   vector<NODE> seg;
71A   vector<TAG> lazy;

249   explicit SegTreeBeats(int n) : N(n), seg(4 * n), lazy(4 * n) {}

67A   template <typename T>
47B   SegTreeBeats(const vector<T> &v)
001       : SegTreeBeats((int)v.size()) {
231       build(1, 0, N - 1, v);
1EF   }

67A   template <typename T>
5B9   void build(int no, int l, int r, const vector<T> &v) {
893       if (l == r) {
C3E         seg[no] = NODE(v[l]);
505         return;
B2E       }
27E       int m = (l + r) >> 1;
35E       build(no << 1, l, m, v);
C55       build((no << 1) | 1, m + 1, r, v);
DEB       seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
349   }

F1A   void push(int no, int lo, int hi) {
946       int m = (lo + hi) >> 1;
8F3       int l = no << 1, r = l | 1;
009       seg[l].apply(lazy[no], lo, m);
9FE       lazy[l].compose(lazy[no]);

204       seg[r].apply(lazy[no], m + 1, hi);
3FC       lazy[r].compose(lazy[no]);

47F       lazy[no] = TAG();
DD1   }

130   void update(int no, int l, int r, int a, int b, const TAG &v) {
F8B       if (b < l || r < a || seg[no].break_condition(v)) return;
0A9       if (l == r) return seg[no].apply(v, l, r), void();
40D       if (a <= l && r <= b && seg[no].tag_condition(v)) {
8C7         seg[no].apply(v, l, r);
92C         lazy[no].compose(v);
505         return;
2B2       }
FF9       push(no, l, r);
27E       int m = (l + r) >> 1;
9AB       update(no << 1, l, m, a, b, v);
087       update((no << 1) | 1, m + 1, r, a, b, v);
DEB       seg[no] = NODE::merge(seg[no << 1], seg[(no << 1) | 1]);
F37   }

38D   NODE query(int no, int l, int r, int a, int b) {
2E6       if (b < l || r < a) return NODE();
83F       if (a <= l && r <= b) return seg[no];
FF9       push(no, l, r);
27E       int m = (l + r) >> 1;
AFE       return NODE::merge(query(no << 1, l, m, a, b),
074                               query((no << 1) | 1, m + 1, r, a, b));
80A   }

4E7   void update(int l, int r, const TAG &v) {
E27       update(1, 0, N - 1, l, r, v);
0D7   }

```

```

36F   NODE query(int l, int r) { return query(1, 0, N - 1, l, r); }
FCC };

```

Dynamic Segment Tree

```

/* Dynamic Segment Tree + Lazy (Range Update, Range Query)
 * Utilizada para intervalos gigantes (ex: 0 a  $10^{18}$ ).
 * Complexidade:  $O(\log N)$  por operacao.
 * Memoria:  $O(Q \log N)$  onde Q e o numero de operacoes.
 * Requisitos:
 * - NODE deve ter: static merge(const NODE&, const NODE&),
 *   apply(const TAG&, ll, ll) e construtor identidade.
 * - TAG deve ter: apply(const TAG&), campo 'has' (bool) e
 *   construtor identidade.
 */

/* --- Exemplo de NODE e TAG ---
struct Tag {
    ll add = 0;
    bool has = false;
    void apply(const Tag& t) {
        add += t.add;
        has = true;
    }
};

struct Node {
    ll val = 0;
    static Node merge(const Node& l, const Node& r) {
        return {l.val + r.val};
    }
    void apply(const Tag& t, ll l, ll r) {
        val += t.add * (r - l + 1);
    }
};
*/

708 template <typename NODE, typename TAG>
77D struct DynamicSegTree {
126   struct InternalNode {
4EB     NODE node;
657     TAG tag;
74F     int l, r;
FB6     InternalNode() : node(NODE()), tag(TAG()), l(-1), r(-1) {}

A61     void apply(const TAG &t, ll l, ll r) {
BE6       node.apply(t, l, r);
D21       tag.apply(t);
9C3     }
2FB   };

983   ll L, R;
4C1   vector<InternalNode> seg;

5C7   DynamicSegTree(ll l, ll r) : L(l), R(r) {
7A0       seg.reserve(4e6);
23E       seg.emplace_back();
BFD   }

4F9   void push(int v, ll l, ll r) {
A16       if (!seg[v].tag.has) return;
56B       if (l == r) return seg[v].tag = TAG();

4D5       if (seg[v].l == -1) {
9EF         seg[v].l = seg.size();
23E         seg.emplace_back();
579       }
CF3       if (seg[v].r == -1) {
90C         seg[v].r = seg.size();
23E         seg.emplace_back();
203       }

769       ll mid = l + (r - l) / 2;
F3E       seg[seg[v].l].apply(seg[v].tag, l, mid);
0B7       seg[seg[v].r].apply(seg[v].tag, mid + 1, r);

DD0       seg[v].tag = TAG();
AD7   }

```

```

352 void update(int v, ll l, ll r, ll ql, ll qr, const TAG &t) {
151     if (ql <= l && r <= qr) return seg[v].apply(t, l, r);
A48     push(v, l, r);
769     ll mid = l + (r - l) / 2;
441     if (ql <= mid) {
405         if (seg[v].l == -1) {
9EF             seg[v].l = seg.size();
23E             seg.emplace_back();
579         }
20D         update(seg[v].l, l, mid, ql, qr, t);
0B1     }
64D     if (qr > mid) {
CF3         if (seg[v].r == -1) {
90C             seg[v].r = seg.size();
23E             seg.emplace_back();
203         }
76B         update(seg[v].r, mid + 1, r, ql, qr, t);
216     }
671     NODE lc = (seg[v].l != -1) ? seg[seg[v].l].node : NODE();
8A0     NODE rc = (seg[v].r != -1) ? seg[seg[v].r].node : NODE();
37A     seg[v].node = NODE::merge(lc, rc);
BF0 }
EE0 NODE query(int v, ll l, ll r, ll ql, ll qr) {
5CF     if (v == -1 || ql > r || qr < l) return NODE();
3C9     if (ql <= l && r <= qr) return seg[v].node;
A48     push(v, l, r);
769     ll mid = l + (r - l) / 2;
211     return NODE::merge(query(seg[v].l, l, mid, ql, qr),
0CC                          query(seg[v].r, mid + 1, r, ql, qr));
933 }
FEE void update(ll ql, ll qr, const TAG &t) {
B79     update(0, L, R, ql, qr, t);
4CA }
323 NODE query(ll ql, ll qr) { return query(0, L, R, ql, qr); }
0BC };

```

Dynamic Persistent Segment Tree

```

/* Persistent Dynamic Segment Tree (Point Update, Range Query)
 * Consulta versoes anteriores e aloca nos sob demanda.
 * Ideal para intervalos grandes e multiplas versoes.
 * Complexidade: O(log N) por update e query.
 * Memoria: O(Q log N) onde Q e o numero de updates.
 * Requisitos:
 * - NODE deve ter: static merge(const NODE&, const NODE&) e
 *   construtor identidade.
 */

/* --- Exemplo de NODE (Soma) ---
struct Node {
    int val;
    Node(int v = 0) : val(v) {}
    static Node merge(const Node& l, const Node& r) {
        return Node(l.val + r.val);
    }
};
*/

8A0 template <typename NODE>
68A struct PersistentSegmentTree {
126     struct InternalNode {
1DF         NODE data;
74F         int l, r;
E61         InternalNode() : data(NODE()), l(0), r(0) {}
B12     };
060     int N;
788     vector<InternalNode> st;
34F     vector<int> roots;
BC9     PersistentSegmentTree(int n) : N(n) {
4E7         st.reserve(4e6);
F96         st.emplace_back();

```

```

405     roots.push_back(0);
71E }
D69 int update(int root, int L, int R, int idx, NODE v) {
4C0     int node = st.size();
DF4     if (root == 0) st.emplace_back();
8FC     else st.push_back({st[root]});
651     if (L == R) {
DC5         st[node].data = NODE::merge(st[node].data, v);
192         return node;
BE8     }
08E     int mid = L + (R - L) / 2;
// A ordem do LHS importa antes do C++17.
D33     if (idx <= mid) {
561         int prev = (root == 0) ? 0 : st[root].l;
339         st[node].l = update(prev, L, mid, idx, v);
413     } else {
AFD         int prev = (root == 0) ? 0 : st[root].r;
674         st[node].r = update(prev, mid + 1, R, idx, v);
066     }
98D     auto& nd = st[node];
0EB     nd.data = NODE::merge(st[nd.l].data, st[nd.r].data);
192     return node;
60F }
082 NODE query(int node, int L, int R, int i, int j) {
0B8     if (node == 0 || i > R || j < L) return NODE();
692     if (i <= L && R <= j) return st[node].data;
08E     int mid = L + (R - L) / 2;
B1E     return NODE::merge(query(st[node].l, L, mid, i, j),
521                          query(st[node].r, mid + 1, R, i, j));
26E }
D6B void update(int idx, NODE v) {
69E     roots.push_back(update(roots.back(), 0, N - 1, idx, v));
1BF }
3A5 NODE query(int version, int l, int r) {
191     return query(roots[version], 0, N - 1, l, r);
247 }
B56 };

```

Line Container (CHT)

```

/* Convex Hull Trick dinamico - O(log N) por operacao
 *
 * Mantem retas y = k·x + m e responde o MAXIMO em x.
 * Aceita k e x arbitrarios; nao exige monotonicidade.
 *
 * Uso: DP dp[i] = max_j(k[j]·x[i] + m[j]). Cada estado vira
 * add(k[j], m[j]); a transicao vira query(x[i]).
 *
 * add(k, m) → insere a reta y = k·x + m
 * query(x) → maior valor entre todas as retas em x
 *
 * Para MÍNIMO: insira add(-k, -m) e use -query(x).
 */

72C struct Line {
F3E     ll k, m;
6F3     mutable ll p;
CA5     bool operator<(const Line &o) const { return k < o.k; }
ABF     bool operator<(ll x) const { return p < x; }
868 };
781 struct LineContainer : multiset<Line, less<>> {
FD2     static const ll inf = LLONG_MAX;
10F     ll div(ll a, ll b) { return a / b - ((a ^ b) < 0 && a % b); }
A1C     bool isect(iterator x, iterator y) {
A95         if (y == end()) return x->p = inf, 0;
9CB         if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
591         else x->p = div(y->m - x->m, x->k - y->k);
870         return x->p >= y->p;
2FA     }
A0C     void add(ll k, ll m) {

```

```

116     auto z = insert({k, m, 0}), y = z++, x = y;
7B1     while (isect(y, z)) z = erase(z);
141     if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
57D     while ((y = x) != begin() && (--x)->p >= y->p)
774         isect(x, erase(y));
086 }
4AD ll query(ll x) {
229     assert(!empty());
7D1     auto l = *lower_bound(x);
96A     return l.k * x + l.m;
D21 }
577 };

```

Graphs

Dijkstra

```

/* Dijkstra - Caminho minimo de fonte unica
 * Complexidade: O((V + E) log V)
 * Requer pesos nao-negativos.
 *
 * adj[u] = lista de {v, w}: aresta u→v com peso w
 *
 * dist[v] = distancia minima de s ate v.
 * dist[v] = INF se v e inalcançavel.
 */
67A template <typename T>
5D5 vector<T> dijkstra(const vector<vector<pair<int, T>>> &adj,
E60                      int s) {
B4C     int n = adj.size();
305     const T INF = numeric_limits<T>::max();
835     vector<T> dist(n, INF);
69C     priority_queue<pair<T, int>, vector<pair<T, int>>, greater<>>
018         pq;
A93     dist[s] = 0;
7BA     pq.push({0, s});
502     while (!pq.empty()) {
2F9         auto [d, u] = pq.top();
716         pq.pop();
3E1         if (d > dist[u]) continue;
610         for (auto [v, w] : adj[u])
DDE             if (dist[u] + w < dist[v]) {
491                 dist[v] = dist[u] + w;
BF3                 pq.push({dist[v], v});
D74             }
9AE         }
8D7     return dist;
6D3 }

```

MST (Kruskal)

```

32D #include "dsu.hpp"

/* MST - Arvore Geradora Minima (Kruskal + DSU)
 * Complexidade: O(E log E). Grafo nao-direcionado ponderado.
 *
 * mst(n, edges) → {custo total, arestas usadas}
 * edges: vetor de {peso, u, v}.
 * Se desconexo, devolve a floresta minima; used tem < n-1.
 *
 * Para arvore MAXIMA, negue os pesos ou ordene decrescente.
 */

67A template <typename T>
FD5 pair<T, vector<pair<int, int>>>
652 mst(int n, vector<tuple<T, int, int>> edges) {
FF1     sort(edges.begin(), edges.end());
370     DSU dsu(n);
C80     T total = 0;
24A     vector<pair<int, int>> used;
DC1     for (auto &[w, u, v] : edges)
5DF         if (dsu.join(u, v)) {
532             total += w;

```

```

12C      used.push_back({u, v});
88B    }
0CF    return {total, used};
290 }

```

Euler Path

```

/* Caminho/Circuito Euleriano - Hierholzer - O(V + E)
*
* Condições de existência:
* Grafo não-direcionado:
*   - Circuito: todos os vértices têm grau par.
*   - Caminho: exatamente dois têm grau ímpar.
* Grafo direcionado:
*   - Circuito: in_degree[v] == out_degree[v] para todo v.
*   - Caminho: um vértice com out-in=1, um com in-out=1.
*
* euler(adj, s) -> caminho com E+1 vértices; vazio se inválido.
* adj[u] = lista de {v, edge_id} para arestas u->v.
* No não-direcionado, insira ambas as direções com o mesmo id.
*/

E48 vector<int> euler(vector<vector<pair<int, int>>> &adj, int s) {
CE9   int edges = 0;
14F   for (auto &v : adj) edges += v.size();
      // Para direcionado, remova a divisão por 2.
2D5   edges /= 2;

B27   vector<bool> used(edges, false);
8E1   vector<int> ptr(adj.size(), 0);
DAB   vector<int> path, stk = {s};

07D   while (!stk.empty()) {
9E9     int v = stk.back();
2C4     bool found = false;
EBE     while (ptr[v] < (int)adj[v].size()) {
FFC       auto [u, eid] = adj[v][ptr[v]++];
AC3       if (!used[eid]) {
C3D         used[eid] = true;
967         stk.push_back(u);
E96         found = true;
C2B         break;
BE7       }
7CB     }
8C1     if (!found) {
80A       path.push_back(v);
518       stk.pop_back();
9D9     }
8D6   }

66D   if ((int)path.size() != edges + 1) return {};
3A9   reverse(path.begin(), path.end());
535   return path;
924 }

```

Lca Sparse Table

```

D7B #include "sparse-table.hpp"

/* LCA (Lowest Common Ancestor) via RMQ
* Ancestral comum mais próximo de dois vértices da árvore.
* Complexidade: O(N log N) no build, O(1) por query.
* Memória: O(N log N)
* Requisitos:
*   - Grafo deve ser uma árvore (conexo e acíclico).
* Métodos:
*   - lca(u, v): Retorna o LCA de u e v.
*   - dist(u, v): Retorna a distância entre u e v.
*   - is_ancestor(u, v): Verifica se u é ancestral de v.
*   - parent(u): Retorna o pai de u.
*/

542 struct LCANode {
43D   int dep, id;
89D   LCANode(int d = 1e9, int pos = -1) : dep(d), id(pos) {}
3E4   static LCANode merge(const LCANode &l, const LCANode &r) {

```

```

DC4     return l.dep < r.dep ? l : r;
0A6   }
008 };

33E struct LCA {
060   int N;
41B   vector<int> first, tour, d, p;
73B   SparseTable<LCANode> st;

DFC   LCA(const vector<vector<int>> &adj, int root = 0)
B9F     : N(adj.size()), first(N), d(N), p(N, -1),
CDD       st(vector<LCANode>()) {

AA1     tour.reserve(2 * N);
52E     vector<LCANode> nodes;
A7F     nodes.reserve(2 * N);

70E     auto dfs = [&](auto self, int u, int par, int dep) -> void {
9A7       p[u] = par;
701       d[u] = dep;
D86       first[u] = tour.size();
FD8       tour.push_back(u);
612       nodes.push_back({dep, (int)tour.size() - 1});
372       for (int v : adj[u]) {
D56         if (v == par) continue;
207         self(self, v, u, dep + 1);
FD8         tour.push_back(u);
612         nodes.push_back({dep, (int)tour.size() - 1});
B41       }
958     };

7D9     dfs(dfs, root, -1, 0);
F2E     st = SparseTable<LCANode>(nodes);
490   }

47F   int parent(int u) { return p[u]; }

310   int lca(int u, int v) {
B63     int l = first[u], r = first[v];
A13     if (l > r) swap(l, r);
369     return tour[st.query(l, r).id];
1E6   }

C11   int dist(int u, int v) {
05C     return d[u] + d[v] - 2 * d[lca(u, v)];
465   }

C22   bool is_ancestor(int u, int v) { return lca(u, v) == u; }
19C };

```

Lca Binary Lifting

```

/* LCA (Lowest Common Ancestor) via Binary Lifting
* Ancestral comum mais próximo de dois vértices da árvore.
* Complexidade: O(N log N) no build, O(log N) por query.
* Memória: O(N log N)
* Requisitos:
*   - Grafo deve ser uma árvore (conexo e acíclico).
* Métodos:
*   - lca(u, v): Retorna o LCA de u e v.
*   - dist(u, v): Retorna a distância entre u e v.
*   - is_ancestor(u, v): Verifica se u é ancestral de v.
*   - parent(u): Retorna o pai de u (up[u][0]).
*   - kth_ancestor(u, k): k-ésimo ancestral; -1 se não existe.
*/

33E struct LCA {
8B7   int N, LOG;
SCE   vector<int> d;
5FE   vector<vector<int>>> up;

DFC   LCA(const vector<vector<int>> &adj, int root = 0)
839     : N(adj.size()), LOG(__lg(N) + 1), d(N),
F1E       up(N, vector<int>(LOG + 1, -1)) {

91B     auto dfs = [&](auto self, int u, int p, int dep) -> void {
701       d[u] = dep;
F1D       up[u][0] = (p == -1) ? u : p;
B16       for (int j = 1; j <= LOG; j++)
4D5         up[u][j] = up[up[u][j - 1]][j - 1];

```

```

372     for (int v : adj[u]) {
730       if (v == p) continue;
207       self(self, v, u, dep + 1);
320     }
3B1   };
7D9   dfs(dfs, root, -1, 0);
5A6 }

579   int parent(int u) { return up[u][0] == u ? -1 : up[u][0]; }
6AE   int kth_ancestor(int u, int k) {
472     if (k > d[u]) return -1;
4C8     for (int j = 0; j <= LOG; j++)
518       if ((k >> j) & 1) u = up[u][j];
03F     return u;
51C   }

310   int lca(int u, int v) {
4F1     if (d[u] < d[v]) swap(u, v);
3B4     int diff = d[u] - d[v];
4C8     for (int j = 0; j <= LOG; j++)
122       if ((diff >> j) & 1) u = up[u][j];
60E     if (u == v) return u;
CAE     for (int j = LOG; j >= 0; j--)
76F       if (up[u][j] != up[v][j]) {
0EF         u = up[u][j];
E4F         v = up[v][j];
D97       }
C15     return up[u][0];
B04   }

C11   int dist(int u, int v) {
05C     return d[u] + d[v] - 2 * d[lca(u, v)];
465   }

C22   bool is_ancestor(int u, int v) { return lca(u, v) == u; }
E8E };

```

Tarjan

```

/* Tarjan SCC (Strongly Connected Components)
* Complexidade: O(V + E)
*
* Campos públicos após build():
* comp[v] = SCC de v, em ordem topológica reversa
* scc[i] = lista de vértices do i-ésimo SCC
* dag = grafo condensado sem arestas duplicadas
* n_scc = número de SCCs
*
* Se há aresta SCC a -> SCC b, então comp[a] > comp[b].
* Para ordem topológica direta, percorra n_scc-1 ... 0.
*/

FA3 struct Tarjan {
058   int n, n_scc = 0;
A41   vector<vector<int>>> g, scc, dag;
F7A   vector<int> comp;

770   Tarjan(int n) : n(n), g(n), comp(n, -1) {}
224   Tarjan(const vector<vector<int>> &adj)
422     : n(adj.size()), g(adj), comp(adj.size(), -1) {
6F2     build();
35F   }

C2B   void add_edge(int u, int v) { g[u].push_back(v); }

0A8   void build() {
E7C     vector<int> low(n), num(n, -1), stk;
C05     vector<bool> on_stk(n, false);
2DC     int timer = 0;

36D     function<void(int)> dfs = [&](int v) {
006       low[v] = num[v] = timer++;
B00       stk.push_back(v);
48C       on_stk[v] = true;
E32       for (int u : g[v]) {
403         if (num[u] == -1) {
512           dfs(u);

```

```

C80     low[v] = min(low[v], low[u]);
F19     } else if (on_stk[u]) low[v] = min(low[v], num[u]);
38B     }
46E     if (low[v] == num[v]) {
861       scc.push_back({});
667       while (true) {
B01         int u = stk.back();
518         stk.pop_back();
635         on_stk[u] = false;
65B         comp[u] = n_scc;
588         scc.back().push_back(u);
163         if (u == v) break;
87F       }
385       n_scc++;
6CD     }
5A5   };

830   for (int i = 0; i < n; i++)
917     if (num[i] == -1) dfs(i);

9E4   dag.assign(n_scc, {});
19E   for (int u = 0; u < n; u++)
4FB     for (int v : g[u])
D67       if (comp[u] != comp[v]) dag[comp[u]].push_back(comp[v]);

404   for (auto &adj : dag) {
324     sort(adj.begin(), adj.end());
942     adj.erase(unique(adj.begin(), adj.end()), adj.end());
80E   }
BA2   }
6D7 };

```

Articulations

```

/* Pontos de Articulação – Tarjan
 * Complexidade: O(V + E)
 * Grafo não-direcionado.
 *
 * Sua remoção aumenta o número de componentes conexas.
 *
 * Campos públicos após build():
 * is_ap[v] = true se v é ponto de articulação
 * get()    = lista dos pontos de articulação
 */

E5A struct Articulations {
1A8   int n;
789   vector<vector<int>> g;
BC9   vector<bool> is_ap;

620   Articulations(int n) : n(n), g(n), is_ap(n, false) {}
6F5   Articulations(const vector<vector<int>> &adj)
FDA     : n(adj.size()), g(adj), is_ap(adj.size(), false) {
6F2     build();
B9D   }

58B   void add_edge(int u, int v) {
F2B     g[u].push_back(v);
4D6     g[v].push_back(u);
FFF   }

0A8   void build() {
AC3     vector<int> num(n, -1), low(n);
2DC     int timer = 0;

CF3     function<void(int, int)> dfs = [&](int v, int par) {
006       low[v] = num[v] = timer++;
E07       int children = 0;
E32       for (int u : g[v]) {
403         if (num[u] == -1) {
C5F           children++;
412           dfs(u, v);
C80           low[v] = min(low[v], low[u]);
369           if (par == -1 && children > 1) is_ap[v] = true;
1B3           if (par != -1 && low[u] >= num[v]) is_ap[v] = true;
709         } else if (u != par) {
2D3           low[v] = min(low[v], num[u]);

```

```

933     }
9F1     }
8CA     };

830     for (int i = 0; i < n; i++)
10B       if (num[i] == -1) dfs(i, -1);
DEE   }

// Lista dos pontos de articulação.
D3D   vector<int> get() const {
02F     vector<int> res;
830     for (int i = 0; i < n; i++)
BA3       if (is_ap[i]) res.push_back(i);
B50     return res;
60A   }
A5D };

```

Bridges

```

/* Pontes (Bridges) – Tarjan
 * Complexidade: O(V + E)
 * Grafo não-direcionado.
 *
 * Uma aresta é ponte se sua remoção desconecta o grafo.
 *
 * Campos públicos após build():
 * bridges = lista de arestas {u, v} que são pontes
 */

D44 struct Bridges {
1A8   int n;
789   vector<vector<int>> g;
0CC   vector<pair<int, int>> bridges;

B0A   Bridges(int n) : n(n), g(n) {}
C65   Bridges(const vector<vector<int>> &adj)
30E     : n(adj.size()), g(adj) {
6F2     build();
5F7   }

58B   void add_edge(int u, int v) {
F2B     g[u].push_back(v);
4D6     g[v].push_back(u);
FFF   }

0A8   void build() {
AC3     vector<int> num(n, -1), low(n);
2DC     int timer = 0;

CF3     function<void(int, int)> dfs = [&](int v, int par) {
006       low[v] = num[v] = timer++;
E32       for (int u : g[v]) {
403         if (num[u] == -1) {
412           dfs(u, v);
C80           low[v] = min(low[v], low[u]);
2FE           if (low[u] > num[v]) bridges.push_back({v, u});
FB4         } else if (u != par) {
2D3           low[v] = min(low[v], num[u]);
E66         }
7A3       }
938     };

830     for (int i = 0; i < n; i++)
10B       if (num[i] == -1) dfs(i, -1);
A6F   }
000 };

```

2-SAT

```

C31 #include "tarjan.hpp"

/* 2-SAT – Complexidade: O(V + E)
 *
 * Literal x: use i para x_i=true, ~i para x_i=false.
 *
 * add_impl(x, y) → x → y e contrapositiva ~y → ~x
 * add_or(x, y)   → x ∨ y
 * add_xor(x, y)  → exatamente um de x, y é true

```

```

* add_iff(x, y) → x ↔ y
* force(x)     → x = true
* solve()      → {satisfatível, ans[]}; ans[i] = 0 ou 1
*/

D9D struct TwoSat {
1A8   int n;
0D7   Tarjan t;
DAB   vector<int> ans;

46B   TwoSat(int n) : n(n), t(2 * n), ans(n, -1) {}

// Converte i em 2*i e ~i em 2*i+1.
5D7   static int node(int x) { return x >= 0 ? 2 * x : -2 * x - 1; }

// implicação x → y (e contrapositiva ~y → ~x)
74A   void add_impl(int x, int y) {
D64     int nx = node(x), ny = node(y);
2B2     t.add_edge(nx, ny);
378     t.add_edge(ny ^ 1, nx ^ 1);
255   }

// x ∨ y
F10   void add_or(int x, int y) { add_impl(~x, y); }

// x XOR y (exatamente um verdadeiro)
487   void add_xor(int x, int y) {
27A     add_or(x, y);
0E8     add_or(~x, ~y);
C02   }

// x ↔ y
1FF   void add_iff(int x, int y) {
2D0     add_impl(x, y);
EA9     add_impl(~x, ~y);
721   }

// força x = true
5F8   void force(int x) { add_impl(~x, x); }

A8E   pair<bool, vector<int>> solve() {
2BD     t.build();
E1C     ans.assign(n, -1);
603     for (int i = 0; i < n; i++) {
EBA       if (t.comp[2 * i] == t.comp[2 * i + 1]) return {false, {}};
049     ans[i] = t.comp[2 * i] < t.comp[2 * i + 1] ? 1 : 0;
EF0   }
997     return {true, ans};
194   }
AC6 };

```

HLD Commutative

```

0FA #include "segment-tree.hpp"

/* HLD Comutativo (Update Pontual)
 * Para operações onde merge(A, B) = merge(B, A).
 */

8A0 template <typename NODE>
403 struct HLD {
577   int n, t;
523   vector<int> p, sz, d, head, pos;
115   SegTree<NODE> st;

AA9   HLD(const vector<vector<int>> &adj, const vector<NODE> &vals,
44C     int root = 0)
F92     : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n),
AC1       st(n) {}

898   vector<vector<int>> g = adj;
227   d[root] = 0, p[root] = root;

94B   auto dfs_sz = [&](auto self, int u) -> void {
267     sz[u] = 1;
839     int best_v = -1, max_sz = -1;
A8C     for (auto &v : g[u]) {
567       if (v == p[u]) continue;
C2C       d[v] = d[u] + 1;
E42       p[v] = u;
033       self(self, v);

```

```

CC3      sz[u] += sz[v];
1F6      if (sz[v] > max_sz) {
F66          max_sz = sz[v];
DD0          best_v = v;
D9D      }
A7E      }
2D5      if (best_v != -1) {
BB6          for (int i = 0; i < (int)g[u].size(); i++) {
F10              if (g[u][i] == best_v && i != 0) {
D76                  swap(g[u][0], g[u][i]);
C2B                  break;
27A              }
CAE          }
07D      };
A11      };
B9B      auto dfs_hld = [&](auto self, int u) -> void {
4C8          pos[u] = t++;
BB6          for (int i = 0; i < (int)g[u].size(); i++) {
06E              int v = g[u][i];
567              if (v == p[u]) continue;
BD7              head[v] = (i == 0 ? head[u] : v);
033              self(self, v);
D08          }
113      };
A8A      dfs_sz(dfs_sz, root);
C07      head[root] = root;
C37      dfs_hld(dfs_hld, root);

759      vector<NODE> base(n);
852      for (int i = 0; i < n; i++) base[pos[i]] = vals[i];
E4E      st = SegTree<NODE>(base);
1E0      }

080      void update(int u, const NODE &val) { st.update(pos[u], val); }

0F8      NODE query_path(int u, int v) {
ADA          NODE res;
0E0          while (head[u] != head[v]) {
44B              if (d[head[u]] > d[head[v]]) swap(u, v);
A0D              res = NODE::merge(res, st.query(pos[head[v]], pos[v]));
025              v = p[head[v]];
41F          }
0E0          if (d[u] > d[v]) swap(u, v);
DC7          return NODE::merge(res, st.query(pos[u], pos[v]));
81B      }

51A      NODE query_subtree(int u) {
0CF          return st.query(pos[u], pos[u] + sz[u] - 1);
747      }
058      };

```

Hld Commutative Lazy

```
064 #include "lazy-segment-tree.hpp"
```

```

/* HLD Comutativo com Lazy Propagation
* Para operacoes onde merge(A, B) = merge(B, A).
* Updates e queries em caminhos e subarvores.
*/

```

```

708 template <typename NODE, typename TAG>
403 struct HLD {
577     int n, t;
523     vector<int> p, sz, d, head, pos;
FFF     LazySegmentTree<NODE, TAG> st;

AA9     HLD(const vector<vector<int>> &adj, const vector<NODE> &vals,
44C         int root = 0)
F92         : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n),
AC1         st(n) {

898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;

94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;

```

```

A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
C2C             d[v] = d[u] + 1;
E42             p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
1F6             if (sz[v] > max_sz) {
F66                 max_sz = sz[v];
DD0                 best_v = v;
D9D             }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {
F10                 if (g[u][i] == best_v && i != 0) {
D76                     swap(g[u][0], g[u][i]);
C2B                     break;
27A                 }
CAE             }
07D         }
A11     };

B9B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
BD7             head[v] = (i == 0 ? head[u] : v);
033             self(self, v);
D08         }
113     };

A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);

759     vector<NODE> base(n);
852     for (int i = 0; i < n; i++) base[pos[i]] = vals[i];
9F8     st = LazySegmentTree<NODE, TAG>(base);
631 }

D7C     void update_path(int u, int v, const TAG &tag) {
0E0         while (head[u] != head[v]) {
44B             if (d[head[u]] > d[head[v]]) swap(u, v);
EC8             st.update(pos[head[v]], pos[v], tag);
025             v = p[head[v]];
DD0         }
0E0         if (d[u] > d[v]) swap(u, v);
07A         st.update(pos[u], pos[v], tag);
656     }

0F8     NODE query_path(int u, int v) {
ADA         NODE res;
0E0         while (head[u] != head[v]) {
44B             if (d[head[u]] > d[head[v]]) swap(u, v);
A0D             res = NODE::merge(res, st.query(pos[head[v]], pos[v]));
025             v = p[head[v]];
41F         }
0E0         if (d[u] > d[v]) swap(u, v);
DC7         return NODE::merge(res, st.query(pos[u], pos[v]));
81B     }

F81     void update_subtree(int u, const TAG &tag) {
675         st.update(pos[u], pos[u] + sz[u] - 1, tag);
20D     }

51A     NODE query_subtree(int u) {
0CF         return st.query(pos[u], pos[u] + sz[u] - 1);
747     }
AA5 };

```

```

0F8     NODE query_path(int u, int v) {
ADA         NODE res;
0E0         while (head[u] != head[v]) {
44B             if (d[head[u]] > d[head[v]]) swap(u, v);
A0D             res = NODE::merge(res, st.query(pos[head[v]], pos[v]));
025             v = p[head[v]];
41F         }
0E0         if (d[u] > d[v]) swap(u, v);
DC7         return NODE::merge(res, st.query(pos[u], pos[v]));
81B     }

F81     void update_subtree(int u, const TAG &tag) {
675         st.update(pos[u], pos[u] + sz[u] - 1, tag);
20D     }

51A     NODE query_subtree(int u) {
0CF         return st.query(pos[u], pos[u] + sz[u] - 1);
747     }
AA5 };

```

Hld Non Commutative

```
0FA #include "segment-tree.hpp"
```

```

/* HLD Nao-Comutativo (Update Pontual)
* Para merge(A, B) != merge(B, A).
* DoubleNode guarda as duas direcoes do merge.

```

```

*/
8A0 template <typename NODE>
0FF struct DoubleNode {
A30     NODE down, up;
F17     DoubleNode() : down(NODE()), up(NODE()) {}
77F     DoubleNode(const NODE &n) : down(n), up(n) {}
31A     DoubleNode(const NODE &d, const NODE &u) : down(d), up(u) {}

25A     static DoubleNode merge(const DoubleNode &l,
553                             const DoubleNode &r) {
7E0         return {NODE::merge(l.down, r.down),
503             NODE::merge(r.up, l.up)};
063     }
FB7 };

8A0 template <typename NODE>
403 struct HLD {
577     int n, t;
256     vector<int> p, sz, d, head, pos, tour;
58B     SegTree<DoubleNode<NODE>> st;

AA9     HLD(const vector<vector<int>> &adj, const vector<NODE> &vals,
44C         int root = 0)
F92         : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n),
007         tour(n), st(n) {

898     vector<vector<int>> g = adj;
227     d[root] = 0, p[root] = root;

94B     auto dfs_sz = [&](auto self, int u) -> void {
267         sz[u] = 1;
839         int best_v = -1, max_sz = -1;
A8C         for (auto &v : g[u]) {
567             if (v == p[u]) continue;
C2C             d[v] = d[u] + 1;
E42             p[v] = u;
033             self(self, v);
CC3             sz[u] += sz[v];
1F6             if (sz[v] > max_sz) {
F66                 max_sz = sz[v];
DD0                 best_v = v;
D9D             }
A7E         }
2D5         if (best_v != -1) {
BB6             for (int i = 0; i < (int)g[u].size(); i++) {
F10                 if (g[u][i] == best_v && i != 0) {
D76                     swap(g[u][0], g[u][i]);
C2B                     break;
27A                 }
CAE             }
07D         }
A11     };

B9B     auto dfs_hld = [&](auto self, int u) -> void {
4C8         pos[u] = t++;
6EA         tour[pos[u]] = u;
BB6         for (int i = 0; i < (int)g[u].size(); i++) {
06E             int v = g[u][i];
567             if (v == p[u]) continue;
BD7             head[v] = (i == 0 ? head[u] : v);
033             self(self, v);
D08         }
54A     };

A8A     dfs_sz(dfs_sz, root);
C07     head[root] = root;
C37     dfs_hld(dfs_hld, root);

9FC     vector<DoubleNode<NODE>> base(n);
830     for (int i = 0; i < n; i++)
9A8         base[pos[i]] = DoubleNode<NODE>(vals[i]);
19C     st = SegTree<DoubleNode<NODE>>(base);
05A }

E1B     void update(int u, const NODE &val) {
353         st.update(pos[u], DoubleNode<NODE>(val));
443     }

```

```

0F8  NODE query_path(int u, int v) {
716  NODE l, r;
0E0  while (head[u] != head[v]) {
BF1  if (d[head[u]] > d[head[v]]) {
0D0  l = NODE::merge(l, st.query(pos[head[u]], pos[u]).up);
139  u = p[head[u]];
A08  } else {
455  r = NODE::merge(st.query(pos[head[v]], pos[v]).down, r);
025  v = p[head[v]];
A09  }
02B  }
152  if (d[u] > d[v]) {
AF0  l = NODE::merge(l, st.query(pos[v], pos[u]).up);
EF6  } else {
5AF  r = NODE::merge(st.query(pos[u], pos[v]).down, r);
F3A  }
79F  return NODE::merge(l, r);
22A  }
51A  NODE query_subtree(int u) {
DE6  return st.query(pos[u], pos[u] + sz[u] - 1).down;
588  }
45D };

```

HLD Non Commutative Lazy

```

064 #include "lazy-segment-tree.hpp"

/* HLD Nao-Comutativo com Lazy Propagation
 * Para merge(A, B) != merge(B, A), como matrizes.
 * DoubleNode guarda as ordens normal e invertida do merge.
 */

8A0 template <typename NODE>
0FF struct DoubleNode {
A30  NODE down, up;

F17  DoubleNode() : down(NODE()), up(NODE()) {}
77F  DoubleNode(const NODE &n) : down(n), up(n) {}
31A  DoubleNode(const NODE &d, const NODE &u) : down(d), up(u) {}

25A  static DoubleNode merge(const DoubleNode &l,
553  const DoubleNode &r) {
7E0  return {NODE::merge(l.down, r.down),
5D3  NODE::merge(r.up, l.up)};
063  }

BBF  template <typename TAG>
3CC  void apply(const TAG &t, int l, int r) {
B08  down.apply(t, l, r);
2BF  up.apply(t, l, r);
8FB  }
204 };

708 template <typename NODE, typename TAG>
403 struct HLD {
577  int n, t;
523  vector<int> p, sz, d, head, pos;
C94  LazySegmentTree<DoubleNode<NODE>, TAG> st;

AA9  HLD(const vector<vector<int>> &adj, const vector<NODE> &vals,
44C  int root = 0)
F92  : n(adj.size()), t(0), p(n), sz(n), d(n), head(n), pos(n),
AC1  st(n) {

898  vector<vector<int>> g = adj;
227  d[root] = 0, p[root] = root;

94B  auto dfs_sz = [&](auto self, int u) -> void {
267  sz[u] = 1;
839  int best_v = -1, max_sz = -1;
A8C  for (auto &v : g[u]) {
567  if (v == p[u]) continue;
C2C  d[v] = d[u] + 1;
E42  p[v] = u;
033  self(self, v);
CC3  sz[u] += sz[v];
1F6  if (sz[v] > max_sz) {

```

```

F66  max_sz = sz[v];
DD0  best_v = v;
}
}
A7E  }
2D5  if (best_v != -1) {
BB6  for (int i = 0; i < (int)g[u].size(); i++) {
F10  if (g[u][i] == best_v && i != 0) {
D76  swap(g[u][0], g[u][i]);
C2B  break;
27A  }
CAE  }
07D  }
A11  };

B9B  auto dfs_hld = [&](auto self, int u) -> void {
4C8  pos[u] = t++;
BB6  for (int i = 0; i < (int)g[u].size(); i++) {
06E  int v = g[u][i];
567  if (v == p[u]) continue;
BD7  head[v] = (i == 0 ? head[u] : v);
033  self(self, v);
D08  }
113  };

A8A  dfs_sz(dfs_sz, root);
C07  head[root] = root;
C37  dfs_hld(dfs_hld, root);

9FC  vector<DoubleNode<NODE>> base(n);
830  for (int i = 0; i < n; i++)
9A8  base[pos[i]] = DoubleNode<NODE>(vals[i]);
54E  st = LazySegmentTree<DoubleNode<NODE>, TAG>(base);
716  }

D7C  void update_path(int u, int v, const TAG &tag) {
0E0  while (head[u] != head[v]) {
BF1  if (d[head[u]] > d[head[v]]) {
D76  st.update(pos[head[u]], pos[u], tag);
139  u = p[head[u]];
D79  } else {
EC8  st.update(pos[head[v]], pos[v], tag);
025  v = p[head[v]];
E8B  }
15B  }
B83  if (d[u] > d[v]) st.update(pos[v], pos[u], tag);
052  else st.update(pos[u], pos[v], tag);
855  }

0F8  NODE query_path(int u, int v) {
716  NODE l, r;
0E0  while (head[u] != head[v]) {
BF1  if (d[head[u]] > d[head[v]]) {
0D0  l = NODE::merge(l, st.query(pos[head[u]], pos[u]).up);
139  u = p[head[u]];
A08  } else {
455  r = NODE::merge(st.query(pos[head[v]], pos[v]).down, r);
025  v = p[head[v]];
A09  }
02B  }
152  if (d[u] > d[v]) {
AF0  l = NODE::merge(l, st.query(pos[v], pos[u]).up);
EF6  } else {
5AF  r = NODE::merge(st.query(pos[u], pos[v]).down, r);
F3A  }
79F  return NODE::merge(l, r);
22A  }

F81  void update_subtree(int u, const TAG &tag) {
675  st.update(pos[u], pos[u] + sz[u] - 1, tag);
20D  }

51A  NODE query_subtree(int u) {
DE6  return st.query(pos[u], pos[u] + sz[u] - 1).down;
588  }
3DA };

```

Dinic

```

/* Dinic (Max Flow)
 * Complexidade: O(V^2E) geral; O(EV) em grafos unitarios.
 * Memoria: O(V + E)

 * Metodos:
 * add_edge(u, v, cap) -> aresta u-v com capacidade cap
 * add_edge(u, v, cap, rcap) -> define capacidade reversa
 * flow(s, t) -> fluxo maximo de s a t
 * min_cut(s) -> lado alcancavel da fonte apos flow()
 */

67A template <typename T>
14D struct Dinic {
E9B  struct Edge {
C63  int to, rev;
4AE  T cap;
001  };

060  int N;
ED3  vector<vector<Edge>> g;
538  vector<int> level, iter;

241  Dinic(int n) : N(n), g(n), level(n), iter(n) {}

D8F  void add_edge(int u, int v, T cap, T rcap = 0) {
25C  g[u].push_back({v, (int)g[v].size(), cap});
ACA  g[v].push_back({u, (int)g[u].size() - 1, rcap});
05B  }

123  bool bfs(int s, int t) {
224  fill(level.begin(), level.end(), -1);
26A  queue<int> q;
EC8  level[s] = 0;
08B  q.push(s);
14D  while (!q.empty()) {
B1E  int v = q.front();
833  q.pop();
24E  for (auto &e : g[v])
466  if (e.cap > 0 && level[e.to] < 0) {
BD7  level[e.to] = level[v] + 1;
6F4  q.push(e.to);
A33  }
FF9  }
FCB  return level[t] >= 0;
56A  }

D20  T dfs(int v, int t, T f) {
704  if (v == t) return f;
6E7  for (int &i = iter[v]; i < (int)g[v].size(); i++) {
63C  Edge &e = g[v][i];
2C2  if (e.cap > 0 && level[v] < level[e.to]) {
054  T d = dfs(e.to, t, min(f, e.cap));
1A3  if (d > 0) {
8FC  e.cap -= d;
772  g[e.to][e.rev].cap += d;
BE2  return d;
18D  }
57D  }
E13  }
BB3  return 0;
182  }

903  T flow(int s, int t) {
19A  T res = 0;
8CE  while (bfs(s, t)) {
736  fill(iter.begin(), iter.end(), 0);
A96  T d;
AE9  while ((d = dfs(s, t, numeric_limits<T>::max())) > 0)
337  res += d;
91B  }
B50  return res;
CE8  }

8B0  vector<bool> min_cut(int s) {
F71  vector<bool> vis(N, false);

```

```

26A queue<int> q;
451 vis[s] = true;
08B q.push(s);
14D while (!q.empty()) {
B1E     int v = q.front();
833     q.pop();
24E     for (auto &e : g[v])
F04         if (e.cap > 0 && !vis[e.to]) {
1E8             vis[e.to] = true;
6F4             q.push(e.to);
A56         }
607     }
C81     return vis;
71F }
993 };

```

Min Cost Max Flow

```

/* Min-Cost Max-Flow (SPFA/Bellman-Ford based)
 * Complexidade: O(VE·flow) no pior caso.
 * Memoria: O(V + E)
 *
 * Metodos:
 * add_edge(u, v, cap, cost) → aresta u→v com cap e custo
 * add_edge(u, v, cap, rcap, cost) → capacidade reversa rcap
 * flow(s, t) → {fluxo maximo, custo minimo} de s a t
 * flow(s, t, limit) → limita o fluxo total a limit
 *
 * Observacoes:
 * - Suporta custos negativos (usa SPFA em vez de Dijkstra).
 * - Sem custos negativos, Dijkstra+potenciais da O(E log V)/aug.
 * - Retorna o menor custo para o fluxo maximo obtido.
 */

```

```

39C template <typename Cap, typename Cost>
6F3 struct MCMF {
E9B     struct Edge {
C63         int to, rev;
2E2         Cap cap;
CB9         Cost cost;
BFF     };
060     int N;
ED3     vector<vector<Edge>> g;
7E6     MCMF(int n) : N(n), g(n) {}
251     void add_edge(int u, int v, Cap cap, Cost cost, Cap rcap = 0) {
C5D         g[u].push_back({v, (int)g[v].size(), cap, cost});
D90         g[v].push_back({u, (int)g[u].size() - 1, rcap, -cost});
200     }
C97     pair<Cap, Cost> flow(int s, int t,
ABA         Cap limit = numeric_limits<Cap>::max()) {
8AC         Cap total_flow = 0;
8F8         Cost total_cost = 0;
DD0         while (limit > 0) {
715             vector<Cost> dist(N, numeric_limits<Cost>::max());
7A2             vector<bool> in_queue(N, false);
FF9             vector<int> prev_node(N, -1), prev_edge(N, -1);
A93             dist[s] = 0;
26A             queue<int> q;
08B             q.push(s);
6FD             in_queue[s] = true;
14D             while (!q.empty()) {
B1E                 int v = q.front();
833                 q.pop();
F19                 in_queue[v] = false;
775                 for (int i = 0; i < (int)g[v].size(); i++) {
027                     auto &e = g[v][i];
AB4                     if (e.cap > 0 && dist[v] + e.cost < dist[e.to]) {
9B1                         dist[e.to] = dist[v] + e.cost;
3C0                         prev_node[e.to] = v;
2E2                         prev_edge[e.to] = i;

```

```

5CA             if (!in_queue[e.to]) {
304                 in_queue[e.to] = true;
6F4                 q.push(e.to);
8A6             }
5E7         }
168     }
A8B }
472     if (dist[t] == numeric_limits<Cost>::max()) break;
B96     Cap pushed = limit;
8CE     for (int v = t; v != s; v = prev_node[v])
31C         pushed = min(pushed, g[prev_node[v]][prev_edge[v]].cap);
879     for (int v = t; v != s; v = prev_node[v]) {
49B         auto &e = g[prev_node[v]][prev_edge[v]];
A04         e.cap -= pushed;
B9A         g[v][e.rev].cap += pushed;
F38     }
3BD     total_flow += pushed;
6CE     total_cost += pushed * dist[t];
3FE     limit -= pushed;
B73 }
570     return {total_flow, total_cost};
2AA }
6F3 };

```

Bipartite Matching

```

0AA #include "dinic.hpp"
/* Bipartite Matching via Dinic
 * Matching maximo em grafo bipartido.
 * Complexidade: O(E * sqrt(V))
 *
 * Vertices esquerda: [0, n), direita: [n, n+m).
 *
 * Campos publicos apos match():
 * size = tamanho do matching maximo
 * match_l[i] = par de i na direita; -1 se livre
 * match_r[j] = par de j na esquerda; -1 se livre
 *
 * Metodos:
 * add_edge(i, j) → aresta entre esquerda i e direita j
 * match() → executa o matching, retorna tamanho
 */

```

```

878 struct BipartiteMatching {
14E     int n, m;
690     Dinic<int> g;
3AE     int S, T;
2AF     vector<int> match_l, match_r;
689     int size = 0;
1FD     BipartiteMatching(int n, int m)
DD3         : n(n), m(m), g(n + m + 2), S(n + m), T(n + m + 1),
779         match_l(n, -1), match_r(m, -1) {
1D3         for (int i = 0; i < n; i++) g.add_edge(S, i, 1);
CC1         for (int j = 0; j < m; j++) g.add_edge(n + j, T, 1);
B60     }
EE4     void add_edge(int i, int j) { g.add_edge(i, n + j, 1); }
274     int match() {
5D7         size = g.flow(S, T);
830         for (int i = 0; i < n; i++)
B83             for (auto &e : g.g[i])
772                 if (e.to != S && e.cap == 0) {
846                     match_l[i] = e.to - n;
78E                     match_r[e.to - n] = i;
FE7                 }
1C6             return size;
B47     }
CFA };

```

Math

Mint

```

/* mint – Inteiro modular
 * Complexidade: O(1) por operacao, O(log mod) para inv()
 *
 * template<typename T, T mod>
 *
 * Uso comum:
 * using mint = Mint<int, 998244353>;
 * using mll = Mint<ll, 998244353LL>;
 *
 * Operacoes: +, -, *, /, ==, !=, <
 * inv() → inverso modular (mod deve ser primo)
 */
2DB template <typename T, T mod>
4D0 struct Mint {
CA0     T v;
926     Mint(T v = 0) : v(((v % mod) + mod) % mod) {}
EBE     Mint operator+(Mint o) const {
07D         return v + o.v >= mod ? v + o.v - mod : v + o.v;
E93     }
774     Mint operator-(Mint o) const {
D65         return v - o.v < 0 ? v - o.v + mod : v - o.v;
66C     }
3B1     Mint operator*(Mint o) const { return (ll)v * o.v % mod; }
5B6     Mint operator/(Mint o) const { return *this * o.inv(); }
96C     Mint &operator+=(Mint o) { return *this = *this + o; }
BB0     Mint &operator-=(Mint o) { return *this = *this - o; }
926     Mint &operator*=(Mint o) { return *this = *this * o; }
E1C     Mint &operator/=(Mint o) { return *this = *this / o; }
8D1     bool operator==(Mint o) const { return v == o.v; }
587     bool operator!=(Mint o) const { return v != o.v; }
A45     explicit operator T() const { return v; }
EC5     Mint inv() const {
FD5         T a = v, b = mod, x = 1, y = 0;
1B4         while (b) {
295             T q = a / b;
617             a -= q * b;
257             swap(a, b);
67E             x -= q * y;
9DD             swap(x, y);
367         }
EA5         return x;
OCF     }
617     friend ostream &operator<<(ostream &os, Mint m) {
B1C         return os << m.v;
09A     }
AAA };

```

Fast Exponentiation

```

/* fexp – Exponenciacao rapida modular
 * Complexidade: O(log b)
 *
 * fexp(a, b, mod) → a^b % mod
 * modinv(a, mod) → a^{-1} % mod; requer mod primo (Fermat)
 *
 * Usa __int128 no produto: seguro para mod ate ~9.2·10¹⁸.
 */
42B ll fexp(ll a, ll b, ll mod) {
67C     a %= mod;
3F8     if (a < 0) a += mod;
CE0     ll res = 1;
C71     for (; b > 0; b >>= 1, a = (__int128)a * a % mod)
D47         if (b & 1) res = (__int128)res * a % mod;
B50     return res;
A29 }

```

```

9E3 ll modinv(ll a, ll mod) {
E74   return fexp(a, mod - 2, mod);
64B }

```

Sieve

```

/* Crivo de Eratostenes
* Complexidade: O(N log log N)
*
* sieve(n) → vetor com todos os primos ate n (inclusive).
*/

705 vector<int> sieve(int n) {
EFA   vector<bool> is_prime(n + 1, true);
FD3   vector<int> primes;
19E   is_prime[0] = is_prime[1] = false;
8A9   for (int p = 2; p <= n; p++) {
29C     if (is_prime[p]) {
BCE       primes.push_back(p);
FAB       if ((ll)p * p <= n)
87F         for (int i = p * p; i <= n; i += p) is_prime[i] = false;
284     }
A20   }
912   return primes;

/* Crivo de menor fator primo (SPF) – O(N log log N) build
* spf[i] = menor primo de i; spf[i] == i indica primo.
* Permite fatorar qualquer x <= n em O(log x).
*
* auto spf = spf_sieve(n);
* factor(x, spf) → pares {primo, expoente} ordenados.
*/

830 vector<int> spf_sieve(int n) {
78F   vector<int> spf(n + 1);
2B8   iota(spf.begin(), spf.end(), 0);
225   for (int i = 2; (ll)i * i <= n; i++)
DD0     if (spf[i] == i)
C0A       for (int j = i * i; j <= n; j += i)
480         if (spf[j] == j) spf[j] = i;
D4F   return spf;
2DC }

4FF vector<pair<int, int>> factor(int x, const vector<int> &spf) {
490   vector<pair<int, int>> f;
40E   while (x > 1) {
239     int p = spf[x], e = 0;
FA7     while (x % p == 0) {
43F       x /= p;
1A3       e++;
E87     }
6BF     f.push_back({p, e});
760   }
ABE   return f;
B2C }

/* Divisores de x fora de ordem – O(d(x)).
* Requer x dentro do crivo usado para gerar spf.
* Aplique sort() ao retorno quando precisar de ordem.
*/

17D vector<int> divisors(int x, const vector<int> &spf) {
FC1   vector<int> divs = {1};
D07   for (auto [p, e] : factor(x, spf)) {
F49     int sz = divs.size(), pk = 1;
6E5     for (int k = 0; k < e; k++) {
DC5       pk *= p;
BA1       for (int i = 0; i < sz; i++) divs.push_back(divs[i] * pk);
FB5     }
261   }
81C   return divs;
C23 }

```

Euler Phi

```

/* Crivo Linear – O(N)
*
* linear_sieve(n) → {primes, phi}
*   primes : todos os primos ate n (inclusive)
*   phi    : phi[i] = φ(i) para i em [0, n]
*/

9A7 pair<vector<int>, vector<int>> linear_sieve(int n) {
303   vector<int> phi(n + 1, 0), primes;
614   vector<bool> composite(n + 1, false);
678   phi[1] = 1;
508   for (int i = 2; i <= n; i++) {
C54     if (!composite[i]) {
E74       primes.push_back(i);
ABD       phi[i] = i - 1;
6DB     }
3B9     for (int p : primes) {
A22       if ((ll)i * p > n) break;
847       composite[i * p] = true;
569       if (i % p == 0) {
8EC         phi[i * p] = phi[i] * p;
C2B         break;
792       } else {
812         phi[i * p] = phi[i] * (p - 1);
988       }
C4A     }
839   }
6EF   return {primes, phi};
415 }

Number Theory

/* GCD / LCM – O(log min(a, b)) */
9C4 ll gcd(ll a, ll b) {
D0B   return b ? gcd(b, a % b) : a;
225 }
026 ll lcm(ll a, ll b) {
8B8   return a / gcd(a, b) * b;
15C }

/* Divisores de n – O(sqrt(N))
* Retorna lista de divisores em ordem crescente.
*/

5FA vector<ll> divisors(ll n) {
17E   vector<ll> lo, hi;
148   for (ll i = 1; i * i <= n; i++) {
C06     if (n % i == 0) {
1EB       lo.push_back(i);
9C3       if (i != n / i) hi.push_back(n / i);
B5B     }
2DE   }
FC1   reverse(hi.begin(), hi.end());
3AF   lo.insert(lo.end(), hi.begin(), hi.end());
253   return lo;
B9C }

/* Fatoracao em primos – O(sqrt(N))
* Retorna lista de {primo, expoente} em ordem crescente.
*/

7E3 vector<pair<ll, int>> factorize(ll n) {
618   vector<pair<ll, int>> factors;
D2E   for (ll p = 2; p * p <= n; p++) {
80A     if (n % p == 0) {
A01       int exp = 0;
03E       while (n % p == 0) {
F4A         n /= p;
666         exp++;
6C9       }
13A       factors.push_back({p, exp});
2E9     }
47D   }
992   if (n > 1) factors.push_back({n, 1});
FA9   return factors;

```

```

92A }

/* CRT – Teorema Chines dos Restos – O(log min(m1, m2))
* Retorna {x, lcm} com x ≡ a1 (mod m1), x ≡ a2 (mod m2).
* Retorna {0, 0} se o sistema for incompativel.
*/

8A9 pair<ll, ll> crt(ll a1, ll m1, ll a2, ll m2) {
8F7   ll g = gcd(m1, m2);
916   if ((a2 - a1) % g != 0) return {0, 0};
931   ll mod = m1 / g * m2;
D0B   ll a = m1 / g, b = m2 / g, x = 1, y = 0;
E58   for (ll ta = a, tb = b; tb; ) {
840     ll q = ta / tb;
4AA     ta -= q * tb;
153     swap(ta, tb);
67E     x -= q * y;
9DD     swap(x, y);
54E   }
581   x = (x % (m2 / g) + m2 / g) % (m2 / g);
A62   ll step = (a2 - a1) / g % (m2 / g) * x % (m2 / g);
5FD   ll result = (a1 + m1 * step) % mod;
1EB   return {(result + mod) % mod, mod};
529 }

/* Fatoracao com primos pre-computados – O(n(sqrt(N)))
* Util apos sieve(sqrt(n)) para fatorar varios numeros.
* primes deve conter todos os primos ate sqrt(n).
*/

165 vector<pair<ll, int>> factorize(ll n, const vector<int> &pr) {
618   vector<pair<ll, int>> factors;
38E   for (int p : pr) {
851     if ((ll)p * p > n) break;
80A     if (n % p == 0) {
A01       int exp = 0;
03E       while (n % p == 0) {
F4A         n /= p;
666         exp++;
6C9       }
13A       factors.push_back({p, exp});
2E9     }
6EA   }
992   if (n > 1) factors.push_back({n, 1});
FA9   return factors;
927 }

```

Miller Rabin

```

/* Miller-Rabin + Pollard Rho
*
* is_prime(n) → O(log² n), deterministico para n < 3.3·10¹⁸
* factorize(n) → O(n^{1/4} log n) esperado; repete fatores
*
* Para fatores distintos, use sort + unique.
*/

DA8 ll mulmod(ll a, ll b, ll m) {
C2A   return (__int128)a * b % m;
7D4 }

0A0 bool is_prime(ll n) {
5A7   if (n < 2) return false;
0A8   if (n == 2 || n == 3 || n == 5 || n == 7) return true;
3CE   if (n % 2 == 0 || n % 3 == 0) return false;
AC7   ll d = n - 1;
898   int r = 0;
5B7   while (d % 2 == 0) d /= 2, r++;
C12   for (ll a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) {
89B     if (a >= n) continue;
8C8     ll x = 1, base = a % n, exp = d;
006     for (; exp > 0; exp >= 1, base = mulmod(base, base, n))
5E6       if (exp & 1) x = mulmod(x, base, n);
8A3     if (x == 1 || x == n - 1) continue;
30D     bool composite = true;
21B     for (int i = 0; i < r - 1; i++) {
F38       x = mulmod(x, x, n);

```

```

A11     if (x == n - 1) {
827         composite = false;
C2B         break;
789     }
4C7 }
4F2 if (composite) return false;
C63 }
8A6 return true;
462 }

711 ll pollard_rho(ll n) {
55A     if (n % 2 == 0) return 2;
ADB     ll x = rand() % (n - 2) + 2, y = x, d = 1;
A4B     ll c = rand() % (n - 1) + 1;
4EB     while (d == 1) {
C17         x = (mulmod(x, x, n) + c) % n;
4EE         y = (mulmod(y, y, n) + c) % n;
4EE         y = (mulmod(y, y, n) + c) % n;
DD8         d = __gcd(abs(x - y), n);
565     }
9EC     return d == n ? pollard_rho(n) : d;
C00 }

6C2 vector<ll> factorize(ll n) {
1B9     if (n == 1) return {};
970     if (is_prime(n)) return {n};
B32     ll d = pollard_rho(n);
8AB     auto l = factorize(d), r = factorize(n / d);
3AF     l.insert(l.end(), r.begin(), r.end());
792     return l;
928 }

```

Matrix

```

/* Matrix – Multiplicacao e exponenciacao de matriz quadrada
 * Complexidade: O(n^3 log k) para M^k
 *
 * Use com mint/mll para operacoes modulares automaticas.
 *
 * Matrix<T> M(n)      → identidade n×n
 * Matrix<T> M(n,0)    → matriz n×n zerada
 * M[i][j]             → acesso direto
 * A * B               → multiplicacao O(n^3)
 * M ^ k               → M^k por exponenciacao rapida
 */

67A template <typename T>
BF3 struct Matrix {
1A8     int n;
C24     vector<vector<T>> mat;

049     Matrix(int n, int identity = 1)
9AE         : n(n), mat(n, vector<T>(n, T(0))) {
340         if (identity)
3BD             for (int i = 0; i < n; i++) mat[i][i] = T(1);
6ED     }

90D     vector<T> &operator[](int i) { return mat[i]; }
699     const vector<T> &operator[](int i) const { return mat[i]; }

AFB     Matrix operator*(const Matrix &rhs) const {
83B         Matrix res(n, 0);
830         for (int i = 0; i < n; i++)
E22             for (int k = 0; k < n; k++)
E64                 if (mat[i][k] != T(0))
F90                     for (int j = 0; j < n; j++)
C55                         res[i][j] = res[i][j] + mat[i][k] * rhs.mat[k][j];
B50         return res;
893     }

348     Matrix operator^(ll b) const {
2DF         Matrix res(n), a = *this;
0EB         for (; b > 0; b >>= 1, a = a * a)
FC2             if (b & 1) res = res * a;
B50         return res;
1B3     }
ACF };

```

Gaussian Elimination

```

/* Eliminacao Gaussiana (RREF) – O(N^2 * M)
 * Funciona com double ou Mint<T, mod>.
 *
 * gauss(A, b) → resolve Ax = b.
 * Retorna {x, true} se unica; false nos demais casos.
 */

67A template <typename T>
5D7 pair<vector<T>, bool> gauss(vector<vector<T>> A, vector<T> b) {
FE3     int n = A.size();
3E0     for (int i = 0; i < n; i++) A[i].push_back(b[i]);
3D7     int m = n + 1, rank = 0;
079     for (int col = 0; col < n && rank < n; col++) {
9D3         int sel = rank;
F97         for (int i = rank + 1; i < n; i++)
E75             if (abs(A[i][col]) > abs(A[sel][col])) sel = i;
9E6         if (!(bool)A[sel][col]) continue;
D8D         swap(A[sel], A[rank]);
B46         T iv = T(1) / A[rank][col];
C24         for (int j = col; j < m; j++) A[rank][j] = A[rank][j] * iv;
603         for (int i = 0; i < n; i++) {
BC7             if (i == rank || !(bool)A[i][col]) continue;
BEF             T f = A[i][col];
69A             for (int j = col; j < m; j++)
E47                 A[i][j] = A[i][j] - f * A[rank][j];
371         }
56E         rank++;
3F2     }
175     if (rank < n) return {{}, false};
A1B     vector<T> x(n);
B95     for (int i = 0; i < n; i++) x[i] = A[i].back();
079     return {x, true};
724 }

```

Linear Basis

```

/* Linear Basis (Base Linear XOR)
 * Complexidade: O(B) por operacao; B = numero de bits.
 * Memoria: O(B)
 *
 * Metodos:
 * insert(x) → true se x e linearmente independente
 * contains(x) → x pertence ao espaco gerado pela base
 * max_xor(x=0) → maior x XOR subconjunto da base
 * min_xor() → menor XOR nao-nulo; 0 se base incompleta
 * size() → numero de elementos linearmente independentes
 * merge(other) → insere os elementos de other
 *
 * Observacoes:
 * - Na forma reduzida, cada pivo aparece em uma so linha.
 * - Para o k-esimo XOR, chame reduce() antes de kth().
 */

538 template <typename T = ll, int B = 63>
F45 struct LinearBasis {
E9C     array<T, B + 1> basis{};
1FC     int sz = 0;

220     bool insert(T x) {
C8C         for (int i = B; i >= 0; i--) {
760             if (!(x >> i) & 1) continue;
617             if (!basis[i]) {
C6C                 basis[i] = x;
EB9                 sz++;
8A6                 return true;
50C             }
6B0             x ^= basis[i];
8A2         }
D1F         return false;
CD7     }

394     bool contains(T x) const {

```

```

C8C         for (int i = B; i >= 0; i--) {
760             if (!(x >> i) & 1) continue;
260             if (!basis[i]) return false;
6B0             x ^= basis[i];
1A9         }
8A6         return true;
189     }

C3B     T max_xor(T x = 0) const {
3CC         for (int i = B; i >= 0; i--) x = max(x, x ^ basis[i]);
EA5         return x;
74B     }

BBB     T min_xor() const {
F56         for (int i = 0; i <= B; i++)
15D             if (basis[i]) return basis[i];
BB3         return 0;
5FC     }

21A     int size() const { return sz; }
AC0     void merge(const LinearBasis &other) {
086         for (int i = B; i >= 0; i--)
D67             if (other.basis[i]) insert(other.basis[i]);
1CA     }

/* Reduz a base: cada pivo aparece em uma unica linha.
 * Necessario antes de kth(); gera os 2^sz XORs distintos.
 */
93A     void reduce() {
C8C         for (int i = B; i >= 0; i--) {
F14             if (!basis[i]) continue;
E36             for (int j = i + 1; j <= B; j++)
7E7                 if ((basis[j] >> i) & 1) basis[j] ^= basis[i];
CA5         }
563     }

/* k-esimo menor XOR, 0-indexado, entre 2^sz subconjuntos.
 * Requer reduce(). k=0 e 0; k=1 e o menor nao-nulo.
 */
0B4     T kth(ll k) const {
E55         vector<T> elems;
F56         for (int i = 0; i <= B; i++)
210             if (basis[i]) elems.push_back(basis[i]);
// Ordem crescente do bit mais significativo.
FAC         if (k >= (1LL << (int)elems.size()))
DAA             return -1; // k fora do range
19A         T res = 0;
687         for (int i = 0; i < (int)elems.size(); i++)
A7C             if ((k >> i) & 1) res ^= elems[i];
B50         return res;
B1A     }
DE8 };

```

Unimodal Search

```

/* Busca unimodal – f deve ser unimodal no intervalo dado.
 *
 * ternary_min/max(lo, hi, f) → inteiros, O(log N)
 * golden_min/max(lo, hi, f, iters=200) → contínuo, O(iters)
 */

// Ternary search para minimo em inteiros [lo, hi]
398 template <typename F>
7B4 ll ternary_min(ll lo, ll hi, F f) {
960     while (hi - lo > 2) {
27E         ll m1 = lo + (hi - lo) / 3;
D61         ll m2 = hi - (hi - lo) / 3;
2EE         if (f(m1) <= f(m2)) hi = m2;
F67         else lo = m1;
794     }
23F     ll best = lo;
AC1     for (ll x = lo + 1; x <= hi; x++)
EB8         if (f(x) < f(best)) best = x;
F26     return best;
67D }

```

```

// Ternary search para maximo em inteiros [lo, hi]
398 template <typename F>
810 ll ternary_max(ll lo, ll hi, F f) {
960 while (hi - lo > 2) {
27E ll m1 = lo + (hi - lo) / 3;
D61 ll m2 = hi - (hi - lo) / 3;
750 if (f(m1) >= f(m2)) hi = m2;
F67 else lo = m1;
F46 }
23F ll best = lo;
AC1 for (ll x = lo + 1; x <= hi; x++)
592 if (f(x) > f(best)) best = x;
F26 return best;
1C7 }

// Golden section search para minimo em continuo [lo, hi]
398 template <typename F>
938 double golden_min(double lo, double hi, F f, int iters = 200) {
84E const double phi = (sqrt(5.0) - 1.0) / 2.0; // ≈ 0.618
0D0 double m1 = hi - phi * (hi - lo);
86C double m2 = lo + phi * (hi - lo);
87F double f1 = f(m1), f2 = f(m2);
5B6 for (int i = 0; i < iters; i++) {
F4D if (f1 < f2) {
82B hi = m2;
3E8 m2 = m1;
068 f2 = f1;
3D2 m1 = hi - phi * (hi - lo);
8B5 f1 = f(m1);
735 } else {
CEC lo = m1;
0DE m1 = m2;
36A f1 = f2;
C3A m2 = lo + phi * (hi - lo);
139 f2 = f(m2);
E4A }
DC3 }
E67 return (lo + hi) / 2.0;
2D2 }

// Golden section search para maximo em continuo [lo, hi]
398 template <typename F>
CEA double golden_max(double lo, double hi, F f, int iters = 200) {
564 return golden_min(
5F7 lo, hi, [&](double x) { return -f(x); }, iters);
2E7 }

```

FFT

```

316 using CD = complex<ld>;

/* FFT – Fast Fourier Transform
* Complexidade: O(N log N)
*
* fft(a) → transforma a no lugar; |a| e potencia de 2
* conv(a, b) → convolucao: c[k] = Σ a[i]*b[k-i]
*
* Precisao: seguro se (Σa²+Σb²)*log₂N < 9·10¹⁴.
* Para coeficientes grandes ou modulo, use fft-mod ou NTT.
*/

B4C void fft(vector<CD> &a) {
820 int n = a.size(), log_n = 31 - __builtin_clz(n);

F82 static vector<complex<long double>> R(2, 1);
6B4 static vector<CD> rt(2, 1);
AD8 for (static int k = 2; k < n; k *= 2) {
411 auto x = polar(1.0L, acos(-1.0L) / k);
9D9 R.resize(n);
335 rt.resize(n);
1D3 for (int i = k; i < 2 * k; i++)
CD4 rt[i] = R[i] = i & 1 ? R[i / 2] * x : R[i / 2];
040 }

808 vector<int> rev(n);
830 for (int i = 0; i < n; i++)

```

```

F9A rev[i] = (rev[i / 2] | (i & 1) << log_n) / 2;
830 for (int i = 0; i < n; i++)
A75 if (i < rev[i]) swap(a[i], a[rev[i]]);

657 for (int k = 1; k < n; k *= 2)
1E5 for (int i = 0; i < n; i += 2 * k)
0C2 for (int j = 0; j < k; j++) {
CD2 auto x = (ld *)&rt[j + k], y = (ld *)&a[i + j + k];
259 CD z(x[0] * y[0] - x[1] * y[1],
45D x[0] * y[1] + x[1] * y[0]);
20A a[i + j + k] = a[i + j] - z;
1B0 a[i + j] += z;
707 }
960 }

17B vector<ld> conv(const vector<ld> &a, const vector<ld> &b) {
F88 if (a.empty() || b.empty()) return {};
BBB vector<ld> res(a.size() + b.size() - 1);
E9A int n = 1 << (32 - __builtin_clz(res.size()));
576 vector<CD> in(n), out(n);
F83 copy(begin(a), end(a), begin(in));
D38 for (int i = 0; i < (int)b.size(); i++) in[i].imag(b[i]);
21A fft(in);
11C for (auto &x : in) x *= x;
830 for (int i = 0; i < n; i++)
5DA out[i] = in[-i & (n - 1)] - conj(in[i]);
3D7 fft(out);
2D6 for (int i = 0; i < (int)res.size(); i++)
A2F res[i] = imag(out[i]) / (4 * n);
B50 return res;
B6D }

```

FFT Mod

```

240 #include "fft.hpp"

/* FFT Mod – Convolucao modular com mod arbitrario
* Complexidade: O(N log N) (~2x mais lento que FFT ou NTT)
*
* convMod<mod>(a, b) → c[k] = (Σ a[i]*b[k-i]) % mod
*
* Entradas devem estar em [0, mod).
* Seguro se N·log₂N·mod < 8.6·10¹⁴.
*
* Sem mod: use cut = 1<<15 e remova os operadores % mod.
*/

3F9 template <int mod>
2A7 vector<ll> convMod(const vector<ll> &a, const vector<ll> &b) {
F88 if (a.empty() || b.empty()) return {};
290 vector<ll> res(a.size() + b.size() - 1);
07A int log_n = 32 - __builtin_clz(res.size());
8E1 int n = 1 << log_n;
// Para alta precisao, use 1<<15 e tire os % mod abaixo.
528 int cut = (int)sqrt(mod);
584 vector<CD> L(n), R(n), outs(n), outl(n);
5B0 for (int i = 0; i < (int)a.size(); i++)
8B7 L[i] = CD((int)a[i] / cut, (int)a[i] % cut);
81B for (int i = 0; i < (int)b.size(); i++)
F55 R[i] = CD((int)b[i] / cut, (int)b[i] % cut);
5D5 fft(L), fft(R);
603 for (int i = 0; i < n; i++) {
39D int j = -i & (n - 1);
65E outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
482 outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / CD(0, 1);
1C6 }
D08 fft(outl), fft(outs);
6D1 for (int i = 0; i < (int)res.size(); i++) {
// alta precisao: tire os % mod nas linhas abaixo
54F ll av = (ll)(real(outl[i]) + .5) % mod;
FA2 ll bv = (ll)(imag(outl[i]) + .5) + (ll)(real(outs[i]) + .5);
A36 ll cv = (ll)(imag(outs[i]) + .5);
557 res[i] = ((av * cut + bv) % mod * cut + cv) % mod;
F04 }
B50 return res;

```

CA2 }

NTT

```

/* NTT – Number Theoretic Transform
* Complexidade: O(N log N)
*
* NTT<mod, g>::conv(a, b) → c[k] = (Σ a[i]*b[k-i]) % mod
*
* Entradas devem estar em [0, mod).
* mod e primo 2^a·b+1; g e sua raiz primitiva.
*
* Opcoes de {mod, g}:
* {998244353, 3} → 119·2²³+1, mais comum, N ≤ 2²³
* {469762049, 3} → 7·2²⁶+1, N ≤ 2²⁶
* {167772161, 3} → 5·2²⁵+1, N ≤ 2²⁵
* {2013265921, 31} → 15·2²⁷+1, N ≤ 2²⁷ (mod > 10⁹)
* {2281701377, 3} → 17·2²⁷+1, N ≤ 2²⁷ (mod > 10⁹)
* {7340033, 5} → 7·2²⁰+1, N ≤ 2²⁰ (menor mod)
* {922337203673735297LL, 3} → 549755813881·2²⁴+1, N≤2²⁴
* (mod~2⁶³, ll)
*/

E34 template <ll mod, ll g>
F12 struct NTT {
D3B static ll fexp(ll a, ll b) {
CE0 ll res = 1;
67C a %= mod;
FF0 for (; b > 0; b >= 1, a = a * a % mod)
602 if (b & 1) res = res * a % mod;
B50 return res;
B0B }

936 static void ntt(vector<ll> &a) {
820 int n = a.size(), log_n = 31 - __builtin_clz(n);
D51 static vector<ll> rt(2, 1);
8EE for (static int k = 2, s = 2; k < n; k *= 2, s++) {
335 rt.resize(n);
277 ll z[] = {1, fexp(g, mod >> s)};
1D3 for (int i = k; i < 2 * k; i++)
256 rt[i] = rt[i / 2] * z[i & 1] % mod;
D00 }
808 vector<int> rev(n);
830 for (int i = 0; i < n; i++)
F9A rev[i] = (rev[i / 2] | (i & 1) << log_n) / 2;
830 for (int i = 0; i < n; i++)
A75 if (i < rev[i]) swap(a[i], a[rev[i]]);
657 for (int k = 1; k < n; k *= 2)
1E5 for (int i = 0; i < n; i += 2 * k)
0C2 for (int j = 0; j < k; j++) {
86E ll z = rt[j + k] * a[i + j + k] % mod, &ai = a[i + j];
598 a[i + j + k] = ai - z + (z > ai ? mod : 0);
4B8 ai += z - (ai + z >= mod ? mod : 0);
D6A }
82E }

46D static vector<ll> conv(vector<ll> a, vector<ll> b) {
F88 if (a.empty() || b.empty()) return {};
5C2 int s = a.size() + b.size() - 1;
DE2 int n = 1 << (32 - __builtin_clz(s));
667 a.resize(n), b.resize(n); ntt(a), ntt(b);
DDC ll inv = fexp(n, mod - 2);
6F8 vector<ll> out(n);
830 for (int i = 0; i < n; i++)
50E out[-i & (n - 1)] = a[i] * b[i] % mod * inv % mod;
EC9 ntt(out);
C20 return {out.begin(), out.begin() + s};
659 }
F65 };

FWHT
/* FWHT – Fast Walsh-Hadamard Transform
* Complexidade: O(N log N)
*

```

```

* Convolucao bitwise: c[k] = Σ a[i]*b[j] onde i op j = k
*
* fwht_conv<'^'>(a, b) → convolucao XOR
* fwht_conv<'|'>(a, b) → convolucao OR
* fwht_conv<'&'>(a, b) → convolucao AND
*
* Ajusta os tamanhos para a proxima potencia de 2.
*/

597 template <char op>
823 void fwht(vector<ll> &a, bool inv = false) {
940     int n = a.size();
980     for (int len = 1; len < n; len *= 2)
EBC         for (int i = 0; i < n; i += 2 * len)
7AB             for (int j = 0; j < len; j++) {
032                 ll u = a[i + j], v = a[i + j + len];
FBD                 if (op == '^') a[i + j] = u + v, a[i + j + len] = u - v;
02F                 if (op == '|') a[i + j + len] = v + (inv ? -u : +u);
729                 if (op == '&') a[i + j] = u + (inv ? -v : +v);
1C4             }
609     if (op == '^' && inv)
F33         for (auto &x : a) x /= n;
643 }

597 template <char op>
E0F vector<ll> fwht_conv(vector<ll> a, vector<ll> b) {
43E     int n = 1;
3B9     while (n < (int)max(a.size(), b.size())) n *= 2;
711     a.resize(n);
DD6     b.resize(n);
883     fwht<op>(a);
671     fwht<op>(b);
34E     for (int i = 0; i < n; i++) a[i] *= b[i];
9D7     fwht<op>(a, true);
3F5     return a;
22D }

```

String

KMP

```

/* KMP (Knuth-Morris-Pratt)
* failure[i] = tamanho do maior prefixo proprio de s[0..i]
* que e tambem sufixo.
* Complexidade: O(N) build, O(N + M) busca.
*
* Aplicacoes:
* - Busca de padrao P em texto T em O(|P| + |T|).
* - Menor periodo da string: N - failure[N-1].
* - Contar ocorrencias de todos os prefixos como sufixos.
*/

090 vector<int> kmp_failure(const string &s) {
163     int n = s.size();
D11     vector<int> f(n, 0);
6F5     for (int i = 1; i < n; i++) {
844         int j = f[i - 1];
424         while (j > 0 && s[i] != s[j]) j = f[j - 1];
04B         if (s[i] == s[j]) j++;
199         f[i] = j;
D74     }
ABE     return f;
8B7 }

// Índices de todas as ocorrencias de pat em txt.
A0D vector<int> kmp_search(const string &pat, const string &txt) {
921     string s = pat + "$" + txt;
3EC     vector<int> f = kmp_failure(s);
11E     int p = pat.size();
927     vector<int> pos;
D6B     for (int i = p + 1; i < (int)s.size(); i++)
06B         if (f[i] == p) pos.push_back(i - 2 * p);
D75     return pos;
F47 }

```

Z Function

```

/* Z-Function
* z[i] = maior prefixo de s que coincide com s[i..].
* z[0] = 0 por convencao.
* Complexidade: O(N) build.
*
* Aplicacoes:
* - Busca P em T: calcule Z de P + "$" + T.
* - Ocorrencias de prefixo k: conte z[i] >= k.
* - Menor periodo: menor p | N com z[p] == N-p.
*/

7AD vector<int> z_function(const string &s) {
163     int n = s.size();
2B1     vector<int> z(n, 0);
A5C     for (int i = 1, l = 0, r = 0; i < n; i++) {
20E         if (i < r) z[i] = min(r - i, z[i - l]);
4ED         while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
B76         if (i + z[i] > r) l = i, r = i + z[i];
3BB     }
070     return z;
197 }

// Índices de todas as ocorrencias de pattern em text.
430 vector<int> z_search(const string &pattern, const string &text) {
365     string s = pattern + "$" + text;
EA6     vector<int> z = z_function(s);
E6D     int p = pattern.size();
FCC     vector<int> positions;
D6B     for (int i = p + 1; i < (int)s.size(); i++)
484         if (z[i] == p) positions.push_back(i - p - 1);
C4C     return positions;
BBF }

```

String Hash

```

/* String Hashing (Single e Double Hash)
* Hash polinomial: build O(N), comparacao O(1).
* Complexidade: O(N) build, O(1) query.
* Memoria: O(N)

*
* SingleHash: um par (MOD, BASE), com baixo risco de colisao.
* DoubleHash: dois pares independentes.
*
* Uso:
*     SingleHash h(s);
*     h.get(l, r); // hash de s[l..r]
*     h.get(l, r) == h.get(l2, r2); // compara substrings
*/

/* --- Single Hash --- */
55A struct SingleHash {
029     static const ll MOD = (1LL << 61) - 1;
56C     static const ll BASE;

F42     vector<ll> h, pw;

30E     static ll mul(ll a, ll b) { return (__uint128_t)a * b % MOD; }

67F     static ll add(ll a, ll b) {
9F4         a += b;
BF2         return a >= MOD ? a - MOD : a;
4E8     }

82B     SingleHash() {}

107     SingleHash(const string &s)
2F6         : h(s.size() + 1, 0), pw(s.size() + 1, 1) {
C46         for (int i = 0; i < (int)s.size(); i++) {
4DB             h[i + 1] = add(mul(h[i], BASE), s[i]);
F0F             pw[i + 1] = mul(pw[i], BASE);
0FA         }
485     }

BA8     ll get(int l, int r) const {
128         return add(h[r + 1], MOD - mul(h[l], pw[r - l + 1]));
358     }

```

```

4EF };

E2C const ll SingleHash::BASE = []() {
862     mt19937_64 rng(
854         chrono::steady_clock::now().time_since_epoch().count());
2E2     return uniform_int_distribution<ll>(1e9,
86D         SingleHash::MOD - 1)(rng);
7E5 }();

/* --- Double Hash --- */
2AF struct DoubleHash {
172     static const ll MOD1 = (1LL << 61) - 1;
D89     static const ll MOD2 = 1e9 + 9;
2BF     static const ll BASE1, BASE2;

23A     vector<ll> h1, h2, pw1, pw2;

780     static ll mul1(ll a, ll b) {
C20         return (__uint128_t)a * b % MOD1;
583     }

876     static ll add1(ll a, ll b) {
9F4         a += b;
B86         return a >= MOD1 ? a - MOD1 : a;
577     }

997     static ll mul2(ll a, ll b) { return (__int128)a * b % MOD2; }
F18     static ll add2(ll a, ll b) {
9F4         a += b;
156         return a >= MOD2 ? a - MOD2 : a;
AC6     }

A3E     DoubleHash() {}

3C9     DoubleHash(const string &s)
01A         : h1(s.size() + 1, 0), h2(s.size() + 1, 0),
4CF         pw1(s.size() + 1, 1), pw2(s.size() + 1, 1) {
C46         for (int i = 0; i < (int)s.size(); i++) {
9F0             h1[i + 1] = add1(mul1(h1[i], BASE1), s[i]);
128             h2[i + 1] = add2(mul2(h2[i], BASE2), s[i]);
562             pw1[i + 1] = mul1(pw1[i], BASE1);
961             pw2[i + 1] = mul2(pw2[i], BASE2);
706         }
366     }

// Par de hashes de s[l..r], inclusive.
C89     pair<ll, ll> get(int l, int r) const {
DC7         ll a = add1(h1[r + 1], MOD1 - mul1(h1[l], pw1[r - l + 1]));
932         ll b = add2(h2[r + 1], MOD2 - mul2(h2[l], pw2[r - l + 1]));
594         return {a, b};
731     }

3DF };

BBD const ll DoubleHash::BASE1 = []() {
862     mt19937_64 rng(
854         chrono::steady_clock::now().time_since_epoch().count());
2E2     return uniform_int_distribution<ll>(1e9,
263         DoubleHash::MOD1 - 1)(rng);
C70 }();

0BD const ll DoubleHash::BASE2 = []() {
862     mt19937_64 rng(
F11         chrono::steady_clock::now().time_since_epoch().count() + 1);
2E2     return uniform_int_distribution<ll>(1e9,
91C         DoubleHash::MOD2 - 1)(rng);
13F }();

Trie

/* Trie (Prefix Tree)
* Insercao e busca de strings em O(|s|).
* Memoria: O(N * ALPHA), N = total de caracteres.
* Alfabeto: [BASE, BASE + ALPHA).
*/

71F template <int ALPHA = 26, char BASE = 'a'>
71A struct Trie {
BF2     struct Node {
C02         int ch[ALPHA];
0F9         int cnt;
302         int end;

```

```

D7F     Node() : cnt(0), end(0) { fill(ch, ch + ALPHA, -1); }
B69 };

F5B vector<Node> t;

8E7 Trie() { t.emplace_back(); }

E7D void insert(const string &s) {
3F3     int node = 0;
B4F     for (char c : s) {
E9F         int id = c - BASE;
19E         if (t[node].ch[id] == -1) {
013             t[node].ch[id] = t.size();
83D             t.emplace_back();
838         }
729         node = t[node].ch[id];
59E         t[node].cnt++;
304     }
879     t[node].end++;
2C4 }

BA9 bool search(const string &s) const {
3F3     int node = 0;
B4F     for (char c : s) {
E9F         int id = c - BASE;
D92         if (t[node].ch[id] == -1) return false;
729         node = t[node].ch[id];
047     }
395     return t[node].end > 0;
3FB }

0B1 int count_prefix(const string &s) const {
3F3     int node = 0;
B4F     for (char c : s) {
E9F         int id = c - BASE;
2AA         if (t[node].ch[id] == -1) return 0;
729         node = t[node].ch[id];
88D     }
D17     return t[node].cnt;
673 }

4E2 int count(const string &s) const {
3F3     int node = 0;
B4F     for (char c : s) {
E9F         int id = c - BASE;
2AA         if (t[node].ch[id] == -1) return 0;
729         node = t[node].ch[id];
88D     }
D51     return t[node].end;
899 }
25C };

```

Manacher

```

/* Manacher's Algorithm (Palindrome Detection)
 * Encontra todos os palindromos em tempo linear.
 * Complexidade: O(N) onde N é o tamanho da string.
 * Memoria: O(N)
 * Funcoes:
 * - manacher(s): maior palindromo centrado em i.
 * - palindrome: testa s[i..j] em O(1).
 * - pal_begin(s): maior palindromo começando em i.
 * - pal_end(s): maior palindromo terminando em i.
 */

67A template <typename T>
44D vector<int> manacher(const T &s) {
18F     int l = 0, r = -1, n = s.size();
FC9     vector<int> d1(n), d2(n);
603     for (int i = 0; i < n; i++) {
821         int k = i > r ? 1 : min(d1[l + r - i], r - i);
61A         while (i + k < n && i - k >= 0 && s[i + k] == s[i - k]) k++;
61E         d1[i] = k--;
9F6         if (i + k > r) l = i - k, r = i + k;
950     }
E03     l = 0, r = -1;

```

```

603     for (int i = 0; i < n; i++) {
1A6         int k = i > r ? 0 : min(d2[l + r - i + 1], r - i + 1);
AC1         k++;
A94         while (i + k <= n && i - k >= 0 && s[i + k] == s[i - k])
AC1             k++;
EAA         d2[i] = --k;
26D         if (i + k - 1 > r) l = i - k, r = i + k - 1;
4FE     }
1B2     vector<int> res(2 * n - 1);
5CB     for (int i = 0; i < n; i++) res[2 * i] = 2 * d1[i] - 1;
891     for (int i = 0; i < n - 1; i++)
748         res[2 * i + 1] = 2 * d2[i + 1];
B50     return res;
85A }

67A template <typename T>
BF0 struct palindrome {
F97     vector<int> man;

B2D     palindrome(const T &s) : man(manacher(s)) {}

1E7     bool query(int i, int j) { return man[i + j] >= j - i + 1; }
933 };

67A template <typename T>
10D vector<int> pal_begin(const T &s) {
542     int n = s.size() - 1;
A23     vector<int> res(s.size());
FDE     palindrome<T> p(s);
371     res[n] = 1;
45B     for (int i = n - 1; i >= 0; i--) {
D9A         res[i] = min(res[i + 1] + 2, n - i + 1);
F2F         while (!p.query(i, i + res[i] - 1)) res[i]--;
DD9     }
B50     return res;
212 }

67A template <typename T>
934 vector<int> pal_end(const T &s) {
A23     vector<int> res(s.size());
FDE     palindrome<T> p(s);
4EC     res[0] = 1;
88E     for (int i = 1; i < s.size(); i++) {
61D         res[i] = min(res[i - 1] + 2, i + 1);
488         while (!p.query(i - res[i] + 1, i)) res[i]--;
5A2     }
B50     return res;
A2C }

```

Suffix Array (NlogN)

```

/* Suffix Array O(N log N) - Prefix Doubling com Count Sort
 * sa[i] = indice do i-esimo sufixo em ordem lexicografica.
 * rk[i] = rank do sufixo s[i..] (inverso de sa).
 * lcp[i] = LCP entre sa[i-1] e sa[i] (lcp[0] = 0).
 * Adiciona '$' ao final da string internamente.
 * Complexidade: O(N log N) build, O(|P| log N) search.
 *
 * Aplicacoes:
 * - Busca de padrao P: search(p) retorna [lo, hi] no SA.
 * - Substring mais longa repetida: *max_element(lcp).
 * - Numero de substrings distintas: N*(N+1)/2 - sum(lcp).
 * - LCS de duas strings: SuffixArray::lcs(s, t).
 */

3F4 struct SuffixArray {
AC0     string s;
58B     vector<int> sa, rk, lcp;

264     SuffixArray() {}

357     SuffixArray(string s) : s(s) {
6F2         build();
E8F         build_lcp();
A87     }

0A8     void build() {
FC8         s += '$';

```

```

163     int n = s.size();
FB9     sa.resize(n);
9BE     rk.resize(n);

413     vector<pair<char, int>> a(n);
490     for (int i = 0; i < n; i++) a[i] = {s[i], i};
3F8     sort(a.begin(), a.end());
67D     for (int i = 0; i < n; i++) sa[i] = a[i].second;
F27     rk[sa[0]] = 0;
6F5     for (int i = 1; i < n; i++) {
7BD         int inc = a[i].first != a[i - 1].first;
6C1         rk[sa[i]] = rk[sa[i - 1]] + inc;
84B     }

8A4     for (int k = 0; (1 << k) < n; k++) {
830         for (int i = 0; i < n; i++)
D82             sa[i] = (sa[i] - (1 << k) + n) % n;
188         count_sort();
037         vector<int> new_rk(n);
CCE         new_rk[sa[0]] = 0;
C6D         auto snd = [&](int x) { return rk[(x + (1 << k)) % n]; };
6F5         for (int i = 1; i < n; i++) {
CD1             pair<int, int> prev = {rk[sa[i - 1]], snd(sa[i - 1])};
3F0             pair<int, int> now = {rk[sa[i]], snd(sa[i])};
AAD             new_rk[sa[i]] = new_rk[sa[i - 1]] + (now != prev);
18C         }
14E         rk = new_rk;
4BE     }
201 }

00C void count_sort() {
9A8     int n = sa.size();
7C4     vector<int> new_sa(n), cnt(n, 0), pos(n, 0);
CBF     for (auto e : rk) cnt[e]++;
3D7     for (int i = 1; i < n; i++) pos[i] = pos[i - 1] + cnt[i - 1];
1EC     for (auto e : sa) new_sa[pos[rk[e]]++] = e;
253     sa = new_sa;
A28 }

47E void build_lcp() {
9A8     int n = sa.size();
75C     lcp.assign(n, 0);
617     for (int i = 0, h = 0; i < n; i++) {
F00         if (rk[i] == 0) {
33D             h = 0;
5E2             continue;
595         }
0A2         int j = sa[rk[i] - 1];
28E         while (s[i + h] == s[j + h]) h++;
CA3         lcp[rk[i]] = h;
254         if (h) h--;
7DF     }
2BE }

// Intervalo [lo, hi] no SA onde p ocorre.
B04 pair<int, int> search(const string &p) const {
B63     int lo, hi, n = sa.size();
F95     {
993         int l = 0, r = n;
40C         while (l < r) {
EE4             int m = (l + r) / 2;
382             if (s.compare(sa[m], p.size(), p) < 0) l = m + 1;
EF3             else r = m;
103         }
0A8         lo = l;
CBD     }
F95     {
993         int l = 0, r = n;
40C         while (l < r) {
EE4             int m = (l + r) / 2;
E02             if (s.compare(sa[m], p.size(), p) <= 0) l = m + 1;
EF3             else r = m;
D29         }
C1B         hi = l;
277     }

```

```

AAF     return {lo, hi};
EEF }

// LCS; sep deve diferir de '$' e nao aparecer em a ou b.
4E7 static string lcs(const string &a, const string &b,
CCC     char sep = 1) {
B29     SuffixArray suf(a + sep + b);
37E     int m = a.size(), best = 0, pos = 0;
564     for (int i = 1; i < (int)suf.sa.size(); i++) {
935         bool in_a = suf.sa[i] < m;
A7D         bool prev_in_a = suf.sa[i - 1] < m;
B09         if (in_a != prev_in_a && suf.lcp[i] > best) {
F6B             best = suf.lcp[i];
2B8             pos = suf.sa[i];
92E         }
409     }
D5D     return suf.s.substr(pos, best);
514 }
019 };

```

Suffix Array (Linear)

```

/* Suffix Array O(N) - SA-IS (Induced Sorting)
* Build O(N). Use quando N muito grande justificar o SA-IS.
*
* Campos publicos apos construcao:
* sa[i] = indice do i-esimo sufixo em ordem lexicografica
* rk[i] = rank do sufixo s[i..] (inverso de sa)
* lcp[i] = tamanho do LCP entre sa[i-1] e sa[i]; lcp[0] = 0
*
* Metodos:
* search(p) -> [lo, hi] no SA; hi-lo e o n° de ocorrencias
*/
FC8 struct SuffixArrayLinear {
AC0     string s;
58B     vector<int> sa, rk, lcp;
CAE     SuffixArrayLinear() {}
B7A     SuffixArrayLinear(const string &s) : s(s + '$') {
C75         vector<int> t(this->s.begin(), this->s.end());
C9A         sa = sa_is(t, 256);
9A8         int n = sa.size();
9BE         rk.resize(n);
665         for (int i = 0; i < n; i++) rk[sa[i]] = i;
E8F         build_lcp();
B40     }

B04     pair<int, int> search(const string &p) const {
B63         int lo, hi, n = sa.size();
F95         {
993             int l = 0, r = n;
40C             while (l < r) {
EE4                 int m = (l + r) / 2;
382                 if (s.compare(sa[m], p.size(), p) < 0) l = m + 1;
EF3                 else r = m;
103             }
0A8             lo = l;
CBD         }
F95         {
993             int l = 0, r = n;
40C             while (l < r) {
EE4                 int m = (l + r) / 2;
E02                 if (s.compare(sa[m], p.size(), p) <= 0) l = m + 1;
EF3                 else r = m;
D29             }
C1B             hi = l;
277         }
AAF         return {lo, hi};
EEF     }

47E     void build_lcp() {
9A8         int n = sa.size();
75C         lcp.assign(n, 0);
E74         for (int i = 0, h = 0; i < n - 1; i++) {

```

```

0A2         int j = sa[rk[i] - 1];
28E         while (s[i + h] == s[j + h]) h++;
CA3         lcp[rk[i]] = h;
254         if (h) h--;
09C     }
C83 }

67A     template <typename T>
EE7     static vector<int> sa_is(const T &s, int sigma) {
163         int n = s.size();
979         if (n == 1) return {0};
4E1         if (n == 2)
32E             return s[0] < s[1] ? vector<int>{0, 1} : vector<int>{1, 0};

81A         vector<bool> t(n);
3A1         t[n - 1] = true;
9C8         for (int i = n - 2; i >= 0; i--)
537             t[i] = s[i] < s[i + 1] || (s[i] == s[i + 1] && t[i + 1]);

A15         auto is_lms = [&](int i) {
BCB             return i > 0 && t[i] && !t[i - 1];
0C4         };

FC6         vector<int> bkt(sigma + 1, 0);
7E5         for (int c : s) bkt[c + 1]++;
CA7         for (int i = 1; i <= sigma; i++) bkt[i] += bkt[i - 1];

CB5         auto induced_sort = [&](const vector<int> &lms) {
386             vector<int> sa(n, -1);
CDA             vector<int> b = bkt;
5EB             for (int i = (int)lms.size() - 1; i >= 0; i--)
F9D                 sa[--b[s[lms[i]] + 1]] = lms[i];
332             b = bkt;
830             for (int i = 0; i < n; i++)
DC3                 if (sa[i] > 0 && !t[sa[i] - 1])
420                     sa[b[s[sa[i] - 1]]++] = sa[i] - 1;
332             b = bkt;
056             for (int i = n - 1; i >= 0; i--)
C0D                 if (sa[i] > 0 && t[sa[i] - 1])
3ED                     sa[--b[s[sa[i] - 1] + 1]] = sa[i] - 1;
DB7             return sa;
66E         };

021         vector<int> lms;
830         for (int i = 0; i < n; i++)
B39             if (is_lms(i)) lms.push_back(i);
EF0         vector<int> sa = induced_sort(lms);

835         vector<int> rk(n, -1);
8F1         int cls = 0;
975         for (int i = 0, prev = -1; i < n; i++) {
FAD             if (!is_lms(sa[i])) continue;
837             if (prev != -1) {
047                 bool diff = false;
5B8                 for (int d = 0;; d++) {
12C                     if (s[prev + d] != s[sa[i] + d] ||
A89                         t[prev + d] != t[sa[i] + d]) {
570                         diff = true;
C2B                         break;
36D                     }
199                     if (d > 0 && (is_lms(prev + d) || is_lms(sa[i] + d)))
C2B                         break;
DA4                 }
265                 if (diff) cls++;
142             }
7F0             rk[sa[i]] = cls;
C6D             prev = sa[i];
9FA         }

CDB         vector<int> lms2, rank2;
830         for (int i = 0; i < n; i++)
6A2             if (rk[i] != -1) {
D24                 lms2.push_back(i);
213                 rank2.push_back(rk[i]);
642             }

// Ha cls+1 classes distintas, numeradas de 0 a cls.

```

```

8F6         vector<int> sa2 = sa_is(rank2, cls + 1);
D14         vector<int> sorted_lms(lms2.size());
E14         for (int i = 0; i < (int)sa2.size(); i++)
42D             sorted_lms[i] = lms2[sa2[i]];

E0A         return induced_sort(sorted_lms);
50A     }
160 };

```

Geometry

Point

```

107 const ld eps = 1e-9;
177 bool eq(ll a, ll b) {
605     return a == b;
6F1 }

D97 bool eq(ld a, ld b) {
BA0     return abs(a - b) <= eps;
BFC }

67A     template <typename T>
F26     struct Point {
645         T x, y;
0FA         Point(T x = 0, T y = 0) : x(x), y(y) {}

C93         bool operator<(Point o) const {
AE1             return tie(x, y) < tie(o.x, o.y);
8D2         }
CE2         bool operator==(Point o) const {
651             return eq(x, o.x) && eq(y, o.y);
511         }
9E5         bool operator!=(Point o) const { return !(*this == o); }

480         Point operator+(Point o) const { return {x + o.x, y + o.y}; }
229         Point operator-(Point o) const { return {x - o.x, y - o.y}; }
AE8         Point operator*(T k) const { return {x * k, y * k}; }
7C2         Point operator/(T k) const { return {x / k, y / k}; }

61A         T dot(Point o) const { return x * o.x + y * o.y; }
CB8         T cross(Point o) const { return x * o.y - y * o.x; }
// area com sinal do triangulo (this, a, b)
DBA         T cross(Point a, Point b) const {
491             return (a - *this).cross(b - *this);
61E         }
F68         T dist2() const { return x * x + y * y; }

AD2         ld len() const { return hypot((ld)x, (ld)y); }
001         Point<ld> norm() const {
DE8             ld l = len();
D97             return {x / l, y / l};
978         }
178         Point perp() const { return {-y, x}; } // rotacao 90° CCW
8CF         Point<ld> rotate(ld a) const { // a em radianos
97E             return {x * cos(a) - y * sin(a), x * sin(a) + y * cos(a)};
CB4         }

539         friend ostream &operator<<(ostream &os, Point p) {
9B9             return os << p.x << " " << p.y;
DB4         }
2F4 };

29C using Pt = Point<ll>;
BA9 using Ptd = Point<ld>;

// angulo entre os vetores p e q
67A     template <typename T>
AE1     ld angle(Point<T> p, Point<T> q) {
9AF         return atan2((ld)p.cross(q), (ld)p.dot(q));
447     }

// area com sinal do triangulo (p, q, r) - positivo = CCW
67A     template <typename T>
D06     T sarea(Point<T> p, Point<T> q, Point<T> r) {
0CC         return p.cross(q, r);
B80     }

// se p, q, r sao colineares

```

```

67A template <typename T>
EB4 bool is_collinear(Point<T> p, Point<T> q, Point<T> r) {
B84     return eq(sarea(p, q, r), (T)0);
0D1 }

Line
C97 #include "point.hpp"

72C struct Line {
2DF     Ptd p, q;
2BD     Line() {}
299     Line(Ptd p, Ptd q) : p(p), q(q) {}
F5B };

// se p pertence ao segmento r
7E1 bool isinseg(Ptd p, Line r) {
AAC     Ptd a = r.p - p, b = r.q - p;
B7E     return eq(a.cross(b), 0) && a.dot(b) < eps;
C3B }

// projecao do ponto p na reta r
15B Ptd proj(Ptd p, Line r) {
BEA     if (r.p == r.q) return r.p;
2FF     Ptd v = r.q - r.p, u = p - r.p;
5C6     return r.p + v * (u.dot(v) / v.dot(v));
2E8 }

// Intersecao das retas; retorna {INF, INF} se paralelas.
24B Ptd inter(Line r, Line s) {
CAD     Ptd u = r.q - r.p, v = s.q - s.p, w = s.p - r.p;
018     ld d = u.cross(v);
F2F     if (eq(d, 0)) return {1e18, 1e18};
2C9     return r.p + u * (w.cross(v) / d);
15F }

// se o segmento r intersecta o segmento s
090 bool interseg(Line r, Line s) {
469     if (isinseg(r.p, s) || isinseg(r.q, s) || isinseg(s.p, r) ||
C1D         isinseg(s.q, r))
8A6         return true;
D02     auto ccw = [] (Ptd a, Ptd b, Ptd c) {
989         return (b - a).cross(c - a) > eps;
3ED     };
13C     return ccw(r.p, r.q, s.p) != ccw(r.p, r.q, s.q) &&
413         ccw(s.p, s.q, r.p) != ccw(s.p, s.q, r.q);
7D4 }

// distancia do ponto p a reta r
E49 ld disttoline(Ptd p, Line r) {
FF0     return abs((r.q - r.p).cross(p - r.p)) / (r.q - r.p).len();
2F5 }

// distancia do ponto p ao segmento r
A20 ld disttoseg(Ptd p, Line r) {
5A8     if ((r.q - r.p).dot(p - r.p) < 0) return (p - r.p).len();
A6E     if ((r.p - r.q).dot(p - r.q) < 0) return (p - r.q).len();
A19     return disttoline(p, r);
9C0 }

// distancia entre dois segmentos
68D ld distseg(Line a, Line b) {
4DF     if (interseg(a, b)) return 0;
3C3     return min({disttoseg(a.p, b), disttoseg(a.q, b),
458         disttoseg(b.p, a), disttoseg(b.q, a)});
A44 }

Circle
5FC #include "line.hpp"

// centro da circunferencia que passa por a, b, c
880 Ptd getcenter(Ptd a, Ptd b, Ptd c) {
3FB     Ptd m1 = (a + b) / 2, m2 = (a + c) / 2;
1E3     return inter(Line(m1, m1 + (b - a).perp()),
9A7         Line(m2, m2 + (c - a).perp()));
E56 }

// intersecao da circunferencia (c, r) com a reta ab

```

```

E6C vector<Ptd> circ_line_inter(Ptd a, Ptd b, Ptd c, ld r) {
5C2     vector<Ptd> points;
FAD     Ptd u = b - a, v = a - c;
F37     ld A = u.dot(u), B = v.dot(u), C = v.dot(v) - r * r;
1FA     ld D = B * B - A * C;
4AA     if (D < -eps) return points;
B40     points.push_back(c + v + u * (-B + sqrt(D + eps)) / A);
AFA     if (D > eps) points.push_back(c + v + u * (-B - sqrt(D)) / A);
97E     return points;
F33 }

// intersecao das circunferencias (a, r) e (b, R)
24A vector<Ptd> circ_inter(Ptd a, Ptd b, ld r, ld R) {
5C2     vector<Ptd> points;
8C7     ld d = (b - a).len();
701     if (d > r + R + eps || d + min(r, R) < max(r, R) - eps)
97E         return points;
398     ld x = (d * d - R * R + r * r) / (2 * d);
A5C     ld y = sqrt(max(0.0, r * r - x * x));
223     Ptd v = (b - a) / d;
EED     points.push_back(a + v * x + v.perp() * y);
526     if (y > eps) points.push_back(a + v * x - v.perp() * y);
97E     return points;
36B }

Polygon
5FC #include "line.hpp"

// 2x a area com sinal (>0 = CCW)
67A template <typename T>
C86 T area2(const vector<Point<T>> &poly) {
D0C     T a = 0;
BC6     int n = poly.size();
830     for (int i = 0; i < n; i++)
F7A         a += poly[i].cross(poly[(i + 1) % n]);
3F5     return a;
1B9 }

// area do poligono
67A template <typename T>
402 ld polarea(const vector<Point<T>> &poly) {
323     return abs((ld)area2(poly)) / 2.0;
3C3 }

// 0 = fora, 1 = dentro, 2 = na borda
// 0(n) para qualquer poligono simples, convexo ou concavo.
67A template <typename T>
EA9 int winding(const vector<Point<T>> &poly, Point<T> p) {
C24     int n = poly.size(), w = 0;
603     for (int i = 0; i < n; i++) {
D79         Point<T> a = poly[i] - p, b = poly[(i + 1) % n] - p;
972         if (a.y <= 0 && b.y > 0) {
E31             T c = a.cross(b);
346             if (c > 0) w++;
EAB             else if (c == 0) return 2;
305         } else if (b.y <= 0 && a.y > 0) {
E31             T c = a.cross(b);
6EA             if (c < 0) w--;
EAB             else if (c == 0) return 2;
992         } else if (b.y == 0 && a.y == 0) {
09D             if (min(a.x, b.x) <= 0 && 0 <= max(a.x, b.x)) return 2;
88E         }
C79     }
2EA     return w != 0 ? 1 : 0;
E31 }

// Ponto no casco convexo CCW sem colineares - O(log n).
67A template <typename T>
406 bool inside_hull(const vector<Point<T>> &hull, Point<T> p) {
ACE     int n = hull.size();
D3E     if (n == 1) return p == hull[0];
4E1     if (n == 2)
DA7         return hull[0].cross(hull[1], p) == 0 &&
540             (p - hull[0]).dot(hull[1] - hull[0]) >= 0 &&
F0B             (p - hull[1]).dot(hull[0] - hull[1]) >= 0;

```

```

D16     if (hull[0].cross(hull[1], p) <= 0 ||
095         hull[0].cross(hull[n - 1], p) >= 0)
D1F         return false;
B9E     int lo = 1, hi = n - 1;
3D1     while (lo + 1 < hi) {
C86         int mid = (lo + hi) / 2;
353         if (hull[0].cross(hull[mid], p) > 0) lo = mid;
8C0         else hi = mid;
575     }
3B9     return hull[lo].cross(hull[lo + 1], p) >= 0;
FF3 }

// Corta pela reta r; mantem p quando (r.p, r.q, p) e CCW.
010 vector<Ptd> cut_polygon(vector<Ptd> poly, Line r) {
2D3     vector<Ptd> res;
BC6     int n = poly.size();
E89     auto ccw = [&] (Ptd a, Ptd b, Ptd c) {
989         return (b - a).cross(c - a) > eps;
2B6     };
603     for (int i = 0; i < n; i++) {
D30         if (ccw(r.p, r.q, poly[i])) res.push_back(poly[i]);
C6F         if (n == 1) continue;
F59         Line s(poly[i], poly[(i + 1) % n]);
DEA         Ptd p = inter(r, s);
D7C         if (isinseg(p, s)) res.push_back(p);
C91     }
E88     res.erase(unique(res.begin(), res.end()), res.end());
EC2     if (res.size() > 1 && res.back() == res[0]) res.pop_back();
B50     return res;
DAD }

// se dois poligonos se intersectam - O(n*m)
B8C bool interpol(const vector<Ptd> &v1, const vector<Ptd> &v2) {
7D1     int n = v1.size(), m = v2.size();
68F     for (auto &p : v1)
E34         if (winding(v2, p)) return true;
043     for (auto &p : v2)
240         if (winding(v1, p)) return true;
830     for (int i = 0; i < n; i++)
A75         for (int j = 0; j < m; j++)
2B6             if (interseg(Line(v1[i], v1[(i + 1) % n]),
DBE                 Line(v2[j], v2[(j + 1) % m])))
8A6                 return true;
D1F     return false;
8F8 }

// Pick: I = (area2 - boundary + 2) / 2, boundary =
// Σgcd(|dx|, |dy|) nas arestas
5CC ll lattice_points(ll area2, ll boundary) {
4EE     return (area2 - boundary + 2) / 2;
4FB }

// distancia entre dois poligonos
CCD ld distpol(const vector<Ptd> &v1, const vector<Ptd> &v2) {
F6B     if (interpol(v1, v2)) return 0;
7D1     int n = v1.size(), m = v2.size();
D63     ld ret = 1e18;
830     for (int i = 0; i < n; i++)
A75         for (int j = 0; j < m; j++)
7DB             ret = min(ret, distseg(Line(v1[i], v1[(i + 1) % n]),
D87                 Line(v2[j], v2[(j + 1) % m])));
EDF     return ret;
B9D }

Convex Hull
046 #include "polygon.hpp"

// Casco convexo CCW, O(N log N); descarta colineares.
67A template <typename T>
580 vector<Point<T>> convex_hull(vector<Point<T>> pts) {
CA0     int n = pts.size();
3C9     if (n < 2) return pts;
CD7     sort(pts.begin(), pts.end());
9C8     vector<Point<T>> h;

```

```

721 h.reserve(2 * n);
107 for (int phase = 0; phase < 2; phase++) {
5C9     int start = h.size();
D9F     for (auto &p : pts) {
C31         while ((int)h.size() >= start + 2) {
531             auto a = h[h.size() - 2], b = h.back();
9F6             if ((b - a).cross(p - a) > 0) break;
BFB             h.pop_back();
0ED         }
A49         h.push_back(p);
37A     }
BFB     h.pop_back();
05C     reverse(pts.begin(), pts.end());
0D0 }
E30 if (h.size() == 2 && h[0] == h[1]) h.pop_back();
81C return h;
5AD }

// Operacoes O(log n) em casco CCW sem colineares.
6E2 struct ConvexPol {
1C4     vector<Ptd> pol;

C45     ConvexPol() {}
D90     ConvexPol(vector<Ptd> v) : pol(convex_hull(v)) {}

// se o ponto esta dentro do casco - O(log n)
46F bool is_inside(Ptd p) {
B1C     int n = pol.size();
03D     if (n == 0) return false;
E08     return inside_hull(pol, p);
66C }

// cmp(p, q) indica que p e mais extremo que q.
8AA int extreme(const function<bool(Ptd, Ptd)> &cmp) {
B1C     int n = pol.size();
4A2     auto extr = [&](int i, bool &cur_dir) {
22A         cur_dir = cmp(pol[(i + 1) % n], pol[i]);
35F         return !cur_dir && !cmp(pol[(i + n - 1) % n], pol[i]);
DDB     };
63D     bool last_dir, cur_dir;
A0D     if (extr(0, last_dir)) return 0;
993     int l = 0, r = n;
EAD     while (l + 1 < r) {
EE4         int m = (l + r) / 2;
F29         if (extr(m, cur_dir)) return m;
44A         bool rel_dir = cmp(pol[m], pol[l]);
72A         if ((!last_dir && cur_dir) ||
D60             (last_dir == cur_dir && rel_dir == cur_dir)) {
8A6             l = m;
1F1             last_dir = cur_dir;
69C         } else r = m;
D0E     }
792     return l;
B61 }

// Índice do ponto com maior produto interno com v - O(log n).
82A int max_dot(Ptd v) {
52C     return extreme(
30C         [&](Ptd p, Ptd q) { return p.dot(v) > q.dot(v); });
C6D }

// Tangentes por p: indices {esquerda, direita} - O(log n).
D16 pair<int, int> tangents(Ptd p) {
2B1     auto left = [&](Ptd q, Ptd r) {
540         return (r - p).cross(q - p) > eps;
3F4     };
183     auto right = [&](Ptd q, Ptd r) {
F28         return (q - p).cross(r - p) > eps;
0FA     };
8EB     return {extreme(left), extreme(right)};
1AB }
A34 };

```

Closest Pair

```
C97 #include "point.hpp"
```

```

/* Closest Pair - Par de pontos mais proximos
 * Complexidade: O(N log N)
 *
 * closest_pair(pts) -> indices {i, j} do par; i < j.
 *
 * Distancia ao quadrado: (pts[i] - pts[j]).dist2()
 * Funciona com coordenadas inteiras, sem erro de precisao.
 */

67A template <typename T>
CCC pair<int, int> closest_pair(const vector<Point<T>> &pts) {
CA0     int n = pts.size();
135     vector<int> idx(n);
295     iota(idx.begin(), idx.end(), 0);
658     sort(idx.begin(), idx.end(),
866         [&](int a, int b) { return pts[a].x < pts[b].x; });

3C9     T best = (pts[idx[0]] - pts[idx[1]]).dist2();
D94     pair<int, int> ans = {idx[0], idx[1]};

// set ordenado por (y, indice)
3B1     set<pair<T, int>> active;
637     for (int l = 0, r = 0; r < n; r++) {
BD5         const Point<T> &pr = pts[idx[r]];
FAF         T delta = (T)ceil(sqrt((double)best)) + 1;
3D6         while (pr.x - pts[idx[l]].x > delta) {
9CB             active.erase({pts[idx[l]].y, idx[l]});
63B             l++;
34B             delta = (T)ceil(sqrt((double)best)) + 1;
B99         }
4A9         auto lo = active.lower_bound({pr.y - delta, INT_MIN});
A87         auto hi = active.upper_bound({pr.y + delta, INT_MAX});
DC0         for (auto it = lo; it != hi; it++) {
6BD             T d = (pr - pts[it->second]).dist2();
1D6             if (d < best) {
365                 best = d;
5A5                 ans = {idx[r], it->second};
C22             }
D09         }
D2B         active.insert({pr.y, idx[r]});
FB8     }
591     if (ans.first > ans.second) swap(ans.first, ans.second);
BA7     return ans;
A71 }

```

Others

Stress Test

```

P=code # codigo a testar
Q=brute # brute correto
C=      # checker; vazio usa cmp

```

```

g++ "$P.cpp" -o sol -O2 || exit 1
g++ "$Q.cpp" -o brt -O2 || exit 1
g++ gen.cpp -o gen -O2 || exit 1
if [ -n "$C" ]; then
    g++ "$C.cpp" -o chk -O2 || exit 1
fi

for ((i = 1; ; i++)); do
    echo "$i"
    ./gen "$i" > in
    ./sol < in > out
    ./brt < in > out2
    if [ -n "$C" ]; then
        ./chk in out out2
    else
        cmp -s out out2
    fi
    if [ $? -ne 0 ]; then

```

```

echo "-> entrada:"
cat in
echo "-> saida code:"
cat out
echo "-> saida brute:"
cat out2
break
fi
done

Gen

42A #define endl '\n'

E72 auto seed =
415     chrono::steady_clock::now().time_since_epoch().count();
F1A     mt19937 rng(seed);
// int x = rng();

463 int uniform(int l, int r) {
A7F     uniform_int_distribution<int> uid(l, r);
F54     return uid(rng);
D9E }

E8D int main() {
886     ios::sync_with_stdio(false);
B95     cin.tie(0);

BB3     return 0;
268 }

```

Checker

```

F3B int main(int argc, char *argv[]) {
4DE     ifstream in(argv[1]), out_sol(argv[2]), out_brt(argv[3]);
BB3     return 0; // 0 = AC, 1 = WA
231 }

```

Hash Function

O hash de cada subseção é calculado com MD5 (3 chars) sobre o código sem comentários e sem espaços em branco. Usado para verificar integridade durante contest.

O hash de uma linha com `{` é o hash de todo o código desde a `{` que a abriu.

Para verificar: copie o código, rode o verificador e compare o prefixo.

Hash Verifier

```

95B string getHash(string code) {
049     ofstream("z.cpp") << code;
FC0     system("g++ -E -P -dD -fpreprocessed ./z.cpp"
041         " | tr -d '[:space:]' | md5sum > sh");
DB3     ifstream("sh") >> code;
78A     return code.substr(0, 3);
497 }

E8D int main() {
E82     string line, context;
0EE     stack<string> blocks({""});
266     while (getline(cin, line)) {
063         context = line;
808         for (char c : line)
66E             if (c == '{') blocks.push("");
CF0             else if (c == '}')
19B                 context = blocks.top() + line, blocks.pop();
661         cout << getHash(context) + " " + line << endl;
4F5         blocks.top() += context;
697     }
56A }

```

Tabelas de Referência

Tipos, Constantes & Complexidade

type	bits	max	fmt	valor	obs
int	32	≈ 2.1 × 10 ⁹	%d	12! = 479001600	int
uint	32	≈ 4.3 × 10 ⁹	%u	20! ≈ 2.4 × 10 ¹⁸	ll
ll	64	≈ 9.2 × 10 ¹⁸	%lld	⁽³³⁾ ₍₁₆₎ ≈ 1.2 × 10 ⁹	int
ull	64	≈ 1.8 × 10 ¹⁹	%llu	⁽⁶⁶⁾ ₍₃₃₎ ≈ 7.2 × 10 ¹⁸	ll

<i>n</i>	π(<i>n</i>)	<i>n</i>	<i>O</i> viavel
10 ⁴	1229	10 ⁸	<i>O</i> (<i>n</i>)
10 ⁵	9592	10 ⁷	<i>O</i> (<i>n</i> log <i>n</i>)
10 ⁶	78498	5 × 10 ³	<i>O</i> (<i>n</i> ²)
10 ⁷	664579	500	<i>O</i> (<i>n</i> ³)
10 ⁸	5761455	22	<i>O</i> (2 ^{<i>n</i>} <i>n</i>)
10 ⁹	50847534	11	<i>O</i> (<i>n</i> !)

Combinatória

Somatórios

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = n^2 \frac{(n+1)^2}{4}$$
$$\sum_{i=1}^n i^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k n^{m+1-k}$$
$$\sum_{i=0}^n c^i = \frac{c^{n+1}-1}{c-1}, \quad \sum_{i=0}^{\infty} c^i = \frac{1}{1-c}, \quad |c| < 1$$
$$\sum_{i=0}^n i c^i = \frac{nc^{n+2} - (n+1)c^{n+1} + c}{(c-1)^2}, \quad \sum_{i=0}^{\infty} i c^i = \frac{c}{(1-c)^2}, \quad |c| < 1$$
$$H_n = \sum_{i=1}^n \frac{1}{i} \approx \ln n + 0.5772, \quad \ln n < H_n < \ln n + 1$$
$$\sum_{i=1}^n H_i = (n+1)H_n - n, \quad \sum_{i=1}^n iH_i = \frac{n(n+1)}{2}H_n - \frac{n(n-1)}{4}$$

Identidades Binomiais

$$\binom{n}{k} = \frac{n!}{(n-k)!k!} = \binom{n}{n-k} = \frac{n}{k} \binom{n-1}{k-1} = \binom{n-1}{k} + \binom{n-1}{k-1}$$
$$\binom{n}{m} \binom{m}{k} = \binom{n}{k} \binom{n-k}{m-k}, \quad \binom{n}{k} = (-1)^k \binom{k-n-1}{k}$$
$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k, \quad x^n - y^n = (x-y) \sum_{k=0}^{n-1} x^{n-1-k} y^k$$
$$\sum_{k=0}^n \binom{n}{k} = 2^n, \quad \sum_{k=0}^n (-1)^k \binom{n}{k} = 0$$
$$\sum_{k=0}^n k \binom{n}{k} = n2^{n-1}, \quad \sum_{k=0}^n k^2 \binom{n}{k} = n(n+1)2^{n-2}$$
$$\sum_{k=0}^n \binom{r+k}{k} = \binom{r+n+1}{n}, \quad \sum_{k=0}^n \binom{k}{m} = \binom{n+1}{m+1}$$
$$\sum_{k=0}^n \binom{r}{k} \binom{s}{n-k} = \binom{r+s}{n}, \quad \sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$$
$$\sum_{k=0}^r (-1)^k \binom{n}{k} = (-1)^r \binom{n-1}{r}$$
$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k, \quad (1+x)^{-n} = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k$$

Inversao binomial: $f(n) = \sum_k \binom{n}{k} g(k) \Leftrightarrow g(n) = \sum_k (-1)^{n-k} \binom{n}{k} f(k)$

Contagem

Permutacoes com repeticoes $n_1,...,n_r$: $\frac{n!}{n_1!\cdots n_r!}$
Permutacoes circulares: $(n-1)!$
Stars & Bars: $x_1 + \cdots + x_k = n, \quad x_i \geq 0$: $\binom{n+k-1}{k-1}$
Inclusao-Exclusao: $|\bigcup_{i=1}^n A_i| = \sum |A_i| - \sum |A_i \cap A_j| + \cdots \pm |A_1 \cap \cdots \cap A_n|$
Sequências Especiais
Catalan: $C_n = \frac{1}{n+1} \binom{2n}{n}$; 1, 1, 2, 5, 14, 42, 132, 429, ...
Conta: arvores binarias com *n* nos, triangulacoes de (*n*+2)-gono, *n* pares de parenteses validos.

Fibonacci: $F_0 = F_1 = 1, \quad F_{-i} = (-1)^{i-1} F_i$
$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}$$
$$F_{i+1}F_{i-1} - F_i^2 = (-1)^i \quad (\text{Cassini})$$
$$F_{n+k} = F_k F_{n+1} + F_{k-1} F_n, \quad E_{2n} = F_n F_{n+1} + F_{n-1} F_n$$
$$\gcd(F_m, F_n) = F_{\gcd(m,n)}, \quad \sum_{k=1}^n F_k = F_{n+2} - 1, \quad \sum_{k=1}^n F_k^2 = F_n F_{n+1}$$
$$\begin{pmatrix} F_n & F_{n-1} \\ F_{n-1} & F_{n-2} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n$$

Representacao de Fibonacci: todo *n* tem representacao unica $n = F_{k_1} + \cdots + F_{k_m}, \quad k_i \geq k_{i+1} + 2, \quad k_m \geq 2.$

Stirling 1ª especie $\begin{bmatrix} n \\ k \end{bmatrix}$: permutacoes de *n* com *k* ciclos.

$$\begin{bmatrix} n \\ 1 \end{bmatrix} = (n-1)!, \quad \begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)! H_{n-1}, \quad \begin{bmatrix} n \\ n \end{bmatrix} = 1$$
$$\begin{bmatrix} n \\ k \end{bmatrix} = (n-1) \begin{bmatrix} n-1 \\ k \end{bmatrix} + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}, \quad \sum_k \begin{bmatrix} n \\ k \end{bmatrix} = n!$$
$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad \begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}$$

Stirling 2ª especie $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$: particoes de *n* em *k* subconjuntos nao vazios.

$$\left\{ \begin{smallmatrix} n \\ 1 \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ n \end{smallmatrix} \right\} = 1, \quad \left\{ \begin{smallmatrix} n \\ 2 \end{smallmatrix} \right\} = 2^{n-1} - 1, \quad \left\{ \begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right\} = \begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}$$
$$\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = k \left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

Bell $B_n = \sum_k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$: # particoes de {1,...,*n*} em subconjuntos nao vazios.
 $B_0 = 1, B_1 = 1, B_2 = 2, B_3 = 5, B_4 = 15, B_5 = 52, B_6 = 203$

Numeros de Euler $\left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle$: permutacoes de *n* com *k* ascendentes.

$$\left\langle \begin{smallmatrix} n \\ 0 \end{smallmatrix} \right\rangle = \left\langle \begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right\rangle = 1, \quad \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle = \left\langle \begin{smallmatrix} n-1 \\ n-k \end{smallmatrix} \right\rangle, \quad \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle = (k+1) \left\langle \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\rangle + (n-k) \left\langle \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\rangle$$
$$\left\langle \begin{smallmatrix} n \\ 1 \end{smallmatrix} \right\rangle = 2^n - n - 1, \quad \left\langle \begin{smallmatrix} n \\ m \end{smallmatrix} \right\rangle = \sum_{k=0}^m (-1)^k \binom{n+1}{k} (m+1-k)^n$$
$$x^n = \sum_{k=0}^n \left\langle \begin{smallmatrix} n \\ k \end{smallmatrix} \right\rangle \binom{x+k}{n}$$

Derangements: $D_n = (n-1)(D_{n-1} + D_{n-2})$; $D_n \approx \frac{n!}{e}$

Josephus: $f(1, k) = 0, \quad f(n, k) = (f(n-1, k) + k) \bmod n$

Funções Geradoras

Serie ordinaria: $A(x) = \sum_{i \geq 0} a_i x^i$; **exponencial:** $A(x) = \sum_{i \geq 0} a_i \frac{x^i}{i!}$
$$\frac{1}{1-x} = \sum x^i, \quad e^x = \sum \frac{x^i}{i!}, \quad \ln(1+x) = \sum (-1)^{i+1} \frac{x^i}{i}$$
$$(1+x)^n = \sum \binom{n}{i} x^i, \quad \frac{x}{1-x-x^2} = \sum F_i x^i$$
$$\frac{1}{2x} (1 - \sqrt{1-4x}) = \sum C_i x^i, \quad \frac{1}{\sqrt{1-4x}} = \sum \binom{2i}{i} x^i$$

Convolucao: $A(x)B(x) = \sum_i \left(\sum_{j=0}^i a_j b_{i-j} \right) x^i.$

Soma acumulada: $B(x) = \frac{1}{1-x} A(x)$ se $b_i = \sum_{j \leq i} a_j.$

Potências Fatoriais & Bernoulli

$$x^{\underline{n}} = x(x-1)\cdots(x-n+1), \quad x^{\overline{n}} = x(x+1)\cdots(x+n-1)$$
$$x^{\underline{n}} = \sum_{k=1}^n \begin{bmatrix} n \\ k \end{bmatrix} (-1)^{n-k} x^k, \quad x^{\overline{n}} = \sum_{k=1}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k$$

$$x^n = \sum_{k=1}^n \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^{\underline{k}}$$

Numeros de Bernoulli ($B_i = 0$ para *i* impar > 1):

$$B_0 = 1, \quad B_1 = -\tfrac{1}{2}, \quad B_2 = \tfrac{1}{6}, \quad B_4 = -\tfrac{1}{30}, \quad B_6 = \tfrac{1}{42}, \quad B_8 = -\tfrac{1}{30}$$
$$\frac{x}{e^x - 1} = \sum_{i \geq 0} B_i \frac{x^i}{i!}, \quad n! = \sqrt{2\pi n} \left(\frac{n}{e} \right)^n \left(1 + \Theta\left(\frac{1}{n}\right) \right)$$

Teoria dos Números

Euler Phi & Möbius

$\varphi(n)$: quantidade de inteiros em [1,*n*] coprimos com *n*.

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p} \right) = \prod_{i=1}^n p_i^{e_i-1} (p_i - 1)$$

$$\varphi(mn) = \varphi(m)\varphi(n) \quad (\gcd = 1), \quad \sum_{d \mid n} \varphi(d) = n$$

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^r n & p_1 \cdots p_r, \quad \sum_{d \mid n} \mu(d) = [n = 1] \end{cases}$$

Inversão de Möbius: $g(n) = \sum_{d \mid n} f(d) \Leftrightarrow f(n) = \sum_{d \mid n} \mu\left(\frac{n}{d}\right) g(d)$

$$\frac{1}{\zeta(x)} = \sum_{i=1}^{\infty} \frac{\mu(i)}{i^x}, \quad \frac{\zeta(x-1)}{\zeta(x)} = \sum_{i=1}^{\infty} \frac{\varphi(i)}{i^x}, \quad \zeta^2(x) = \sum_{i=1}^{\infty} \frac{d(i)}{i^x}$$

GCD, Diofantinas & TCR

$\gcd(a, b) = \gcd(b, a \bmod b)$; $\text{lcm}(a, b) = ab / \gcd(a, b)$

Bezout: $ax + by = \gcd(a, b)$ sempre tem solução inteira. Em geral, $ax + by = c$ tem solução $\Leftrightarrow \gcd(a, b) \mid c.$

TCR: Sistema $x \equiv a_i \pmod{m_i}$ com *m_i* coprimos dois a dois tem solucao unica $\bmod M = \prod m_i$:

$$x = \sum_i a_i M_i (M_i^{-1} \bmod m_i), \quad M_i = \frac{M}{m_i}$$

Aritmetica Modular

Inverso modular: $a^{-1} \bmod m$ via $a^{m-2} \bmod m$ (requer *m* primo). Para todos os inversos 1..*n* de uma vez:

$$\text{inv}[1] = 1, \quad \text{inv}[i] = -(m/i) \cdot \text{inv}[m \bmod i] \bmod m$$

Lucas: Para *p* primo, $\binom{n}{k} \bmod p = \binom{n \bmod p}{k \bmod p} \cdot \binom{\lfloor \frac{n}{p} \rfloor}{\lfloor \frac{k}{p} \rfloor} \pmod{p}$

Recorrências & Séries

Teorema Mestre

$$T(n) = aT\left(\frac{n}{b}\right) + f(n), \quad a \geq 1, \quad b > 1:$$
$$f(n) = O(n^{\log_b a - \varepsilon}) \Rightarrow T(n) = \Theta(n^{\log_b a})$$
$$f(n) = \Theta(n^{\log_b a}) \Rightarrow T(n) = \Theta(n^{\log_b a} \log n)$$
$$f(n) = \Omega(n^{\log_b a + \varepsilon}) \text{ e } af\left(\frac{n}{b}\right) \leq cf(n) \Rightarrow T(n) = \Theta(f(n))$$

Séries de Taylor

$$e^x = \sum \frac{x^i}{i!}, \quad \ln(1+x) = \sum (-1)^{i+1} \frac{x^i}{i}, \quad \ln \frac{1}{1-x} = \sum \frac{x^i}{i}$$
$$\sin x = \sum (-1)^i \frac{x^{2i+1}}{(2i+1)!}, \quad \cos x = \sum (-1)^i \frac{x^{2i}}{(2i)!}$$
$$\arctan x = \sum (-1)^i \frac{x^{2i+1}}{2i+1}, \quad \frac{1}{(1-x)^{n+1}} = \sum \binom{i+n}{i} x^i$$

Teoria dos Grafos

Teoremas

Euler: Circuito Euleriano $\Leftrightarrow$ todos os vertices tem grau par. Caminho Euleriano $\Leftrightarrow$ exatamente 2 vertices de grau impar.

Fórmula de Euler (planar): $V - E + F = 2.$ Grafo planar simples: $E \leq 3V - 6$; planar bipartido: $E \leq 2V - 4.$ Todo planar tem algum vertice de grau $\leq 5.$

Cayley: Número de árvores geradoras rotuladas de *K_n* e *nⁿ⁻²*.

Kirchhoff: Número de árvores geradoras = qualquer cofator da matriz Laplaciana $L = D - A.$

Konig: Em grafo bipartido: cobertura mínima de vértices = emparelhamento máximo; conjunto independente máximo = $V -$ emparelhamento máximo.

Hall: Existe emparelhamento perfeito de $U \Leftrightarrow \forall S \subseteq U: |N(S)| \geq |S|$.

Menger: Número máximo de caminhos aresta-disjuntos entre s e t = tamanho do corte mínimo $s-t$.

Dilworth: Em poset, partição mínima em cadeias = anti-cadeia máxima.

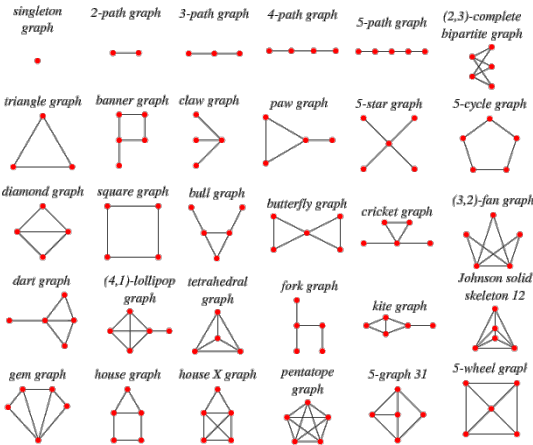
Mirsky: Em poset, partição mínima em anti-cadeias = comprimento da maior cadeia.

Prüfer: Bijecao entre árvores rotuladas de n vértices e seqüências de comprimento $n-2$ sobre $\{1,...,n\}$.

Erdos-Gallai: Seqüência $d_1 \geq ... \geq d_n$ é seqüência de graus de algum grafo simples $\Leftrightarrow \sum d_i$ e par e para todo k :

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$$

Tipos de Grafo



Matemática

Probabilidade

Bayes: Atualiza a probabilidade de A dado que B ocorreu.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Quando $P(B)$ nao e conhecida diretamente (A_j sao hipoteses mutuamente exclusivas e exaustivas):

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$

Esperanca: $E[X] = \sum xP(X = x)$

Linearidade: $E[aX + bY] = aE[X] + bE[Y]$. Se independentes: $E[XY] = E[X]E[Y]$

Variancia: $\text{Var}(X) = E[X^2] - E[X]^2$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Se independentes: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Markov: $P[X \geq \lambda E[X]] \leq \frac{1}{\lambda}$

Chebyshev: $P[|X - E[X]| \geq \lambda \sigma] \leq \frac{1}{\lambda^2}$

Binomial $B(n, p)$: n ensaios de Bernoulli independentes com prob p .

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad E[X] = np, \quad \text{Var} = np(1 - p)$$

Geometrica $\text{Geom}(p)$: numero de tentativas ate o primeiro sucesso ($q = 1 - p$).

$$P(X = k) = pq^{k-1}, \quad E[X] = \frac{1}{p}, \quad \text{Var} = \frac{q}{p^2}$$

Hipergeometrica $H(N, K, n)$: N itens, K marcados, n sorteados sem reposicao.

$$P(X = k) = \binom{K}{k} \frac{\binom{N-K}{n-k}}{\binom{N}{n}}, \quad E[X] = \frac{nK}{N}$$

Poisson $\text{Pois}(\lambda)$: aproximacao de $B(n, p)$ quando $n \rightarrow \infty, np = \lambda$.

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad E[X] = \text{Var}(X) = \lambda$$

Normal $\mathcal{N}(\mu, \sigma^2)$: $p(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad E[X] = \mu, \quad \text{Var} = \sigma^2$.

Coupon collector: n tipos equiprovaveis; numero esperado de sorteios = nH_n .

Paradoxo do aniversario: $\approx 1.18\sqrt{M}$ amostras de $[M]$ para colisao com prob $\geq 50\%$.

Trigonometria

$$\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

$$\sin 2x = 2 \sin x \cos x, \quad \cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$e^{ix} = \cos x + i \sin x, \quad e^{ix} = -1$$

$$\text{Lei dos senos: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\text{Lei dos cossenos: } c^2 = a^2 + b^2 - 2ab \cos C$$

Area do triangulo (lados a,b,c; angulos A,B,C):

$$\frac{1}{2} ab \sin C, \quad \sqrt{s(s-a)(s-b)(s-c)} \quad s = \frac{a+b+c}{2}, \quad \frac{c^2 \sin A \sin B}{2 \sin C}$$

$$\text{Circumraio: } R = ab \frac{c}{4A}, \quad \text{Inraio: } r = \frac{A}{s}.$$

Geometria

$$\text{Shoelace: } A = \frac{1}{2} |\sum_i (x_i y_{i+1} - x_{i+1} y_i)|$$

$$\text{Pick: } A = I + \frac{B}{2} - 1 \quad (I = \text{pontos interiores}, B = \text{pontos na borda})$$

$$\text{Area por coordenadas: } \frac{1}{2} |\det \begin{pmatrix} x_1 - x_3 & x_2 - x_3 \\ y_1 - y_3 & y_2 - y_3 \end{pmatrix}|$$

$$\text{Esferas/circulos: } A = \pi r^2, \quad V = \frac{4}{3} \pi r^3.$$

$$\text{Produto vetorial: } \vec{u} \times \vec{v} = (u_y v_z - u_z v_y, u_z v_x - u_x v_z, u_x v_y - u_y v_x)$$

Em 2D: $u_x v_y - u_y v_x$ (> 0 anti-horario, $= 0$ colinear; modulo = area do paralelogramo)

$$\text{Dist. ponto-ponto: } d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \quad (3D: \text{inclui } (z_1 - z_2)^2)$$

$$\text{Dist. ponto-reta } (ax + by + c = 0): d = |ax_0 + by_0 + c| \frac{1}{\sqrt{a^2 + b^2}}$$

Rotacao 2D:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Rotacao 3D em torno de x, y, z por θ ($c = \cos \theta, s = \sin \theta$):

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{pmatrix}, \quad R_y = \begin{pmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{pmatrix}, \quad R_z = \begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Teoria dos Jogos

Sprague-Grundy: Todo jogo imparcial equivale a um Nim de um unico monte. O valor de Grundy (nimber) de uma posicao u e:

$$g(u) = \text{mex}\{g(v) : v \text{ alcançável de } u\}$$

onde $\text{mex}(S)$ e o menor inteiro nao negativo fora de S .

Posição perdedora $\Leftrightarrow g(u) = 0$. Em jogo composto de partes independentes:

$$g = g_1 \oplus g_2 \oplus \dots \oplus g_k$$

Nim: k montes de tamanhos $a_1, ..., a_k$. Posição perdedora $\Leftrightarrow a_1 \oplus \dots \oplus a_k = 0$.

Nim misere (quem pegar o ultimo *perde*): subtraia 1 de cada pilha; se o XOR resultante $\neq 0$, a posicao original e ganhadora.

Wythoff: 2 montes $(a, b), a \leq b$. Movimentos: remover qualquer qtd de um monte, ou a mesma qtd dos dois. Posição perdedora $\Leftrightarrow a = \lfloor \varphi k \rfloor, b = \lfloor \varphi^2 k \rfloor$ para algum $k \geq 0$, onde $\varphi = \frac{1+\sqrt{5}}{2}$.

Tabelas Numéricas

Números Primos

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73
79 83 89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181
191 193 197 199 211 223 227 229 233 239 241 251 257 263 269 271 277 281 283 293 307
311 313 317 331 337 347 349 353 359 367 373 379 383 389 397 401 409 419 421 431 433
439 443 449 457 461 463 467 479 487 491 499 503 509 521 523 541 547 557 563 569 571
577 587 593 599 601 607 613 617 619 631 641 643 647 653 659 661 673 677 683 691 701
709 719 727 733 739 743 751 757 761 769 773 787 797 809 811 821 823 827 829 839 853
857 859 863 877 881 883 887 907 911 919 929 937 941 947 953 967 971 977 983 991 997

Fibonacci

n	F_n	n	F_n	n	F_n	n	F_n	n	F_n	n	F_n
0	0	1	1	2	1	3	2	4	3	5	5
6	8	7	13	8	21	9	34	10	55	11	89
12	144	13	233	14	377	15	610	16	987	17	1597
18	2584	19	4181	20	6765						

Triângulo de Pascal ($\binom{n}{k}$)

$n \backslash k$	0	1	2	3	4	5	6	7	8	9	10
0	1										
1	1	1									
2	1	2	1								
3	1	3	3	1							
4	1	4	6	4	1						
5	1	5	10	10	5	1					
6	1	6	15	20	15	6	1				
7	1	7	21	35	35	21	7	1			
8	1	8	28	56	70	56	28	8	1		
9	1	9	36	84	126	126	84	36	9	1	
10	1	10	45	120	210	252	210	120	45	10	1

Debugging

```
-g++ -O2 -std=c++17 -Wall -Wextra  
-fsanitize=address,undefined  
-fno-sanitize-recover=all -g
```

Antes de submeter:

- Escreva casos simples se o sample não for suficiente.
- Limite de tempo apertado? Gere casos máximos.
- Memória está ok?
- Algo pode dar overflow?
- Está submetendo o arquivo certo?

Wrong Answer:

- Imprima sua solução e outputs de debug.
- Está limpando estruturas entre casos de teste?
- O algoritmo cobre todo o range de entrada?
- Releia o enunciado completo.
- Todos os casos de borda estão corretos?
- Entendeu o problema corretamente?
- Variáveis não inicializadas?
- Overflow?
- Confundindo N/M, i/j, etc.?
- Tem certeza que o algoritmo funciona?
- Que casos especiais você não considerou?
- As funções da STL funcionam como você acha?
- Adicione asserts e resubmeta.
- Rode em casos criados manualmente.
- Execute o algoritmo num caso simples à mão.
- Explique o algoritmo para um colega.
- Peça para o colega revisar seu código.
- Dê uma volta, vá ao banheiro.
- O formato de saída está correto? (espaços, newlines)
- Reescreva do zero ou peça ao colega.

Runtime Error:

- Testou todos os casos de borda localmente?
- Variáveis não inicializadas?
- Lendo/escrevendo fora do range de algum vetor?
- Algum assert que pode falhar?
- Divisão por zero? (mod 0?)
- Recursão infinita possível?
- Ponteiros ou iteradores invalidados?
- Usando memória demais?

Time Limit Exceeded:

- Tem loop infinito?
- Qual é a complexidade do algoritmo?
- Está copiando dados desnecessariamente? (use referências)
- Tamanho da entrada/saída? (considere scanf)
- Evite map/vector; use arrays/unordered_map.
- O que seus colegas acham do algoritmo?

Memory Limit Exceeded:

- Quanto de memória seu algoritmo realmente precisa?
- Está limpando estruturas entre casos de teste?